AL RYADA UNIVERSITY FOR SCIENCE AND TECHNOLOGY

FACULTY OF NURSING

Critical Care Nursing Clinical

Part 2

Prepared by

Critical Care Nursing Department

University Vision

Al-Ryada University for Science and Technology aspires to be a distinguished and highly competitive leading university at the local and international levels in line with the sustainable development goals.

University Mission

The University is committed to providing a distinguished graduate capable of innovating and competing in the labour market and meeting the challenges of the future by providing a stimulating educational environment with intelligent and sophisticated techniques and strengthening partnerships with local and international universities and institutions and employing scientific research to achieve academic and research excellence while adhering to national identity, ethical values and community responsibility.

Vision of the College:

The College of Nursing at the Al-Ryada University for Science and Technology in Egypt aspires to become a leading college in embracing comprehensive, distinguished education that supports students' personal capabilities and leads to self-confidence and continuous self-development. This positively impacts fair professional competition locally, regionally, and internationally, and advances the educational, research, and societal aspects of the profession.

Mission of the College:

The College of Nursing at the Al-Ryada University for Science and Technology in Egypt is committed to utilizing a variety of modern educational methods and theories and continually developing them in line with scientific developments and educational quality standards. This will result in the graduation of distinguished cadres with a high competitiveness at the local, regional, and international levels in the practice of the nursing profession and in the field of scientific research. The College is committed to actively participating in the continuous development of various aspects of the profession, which will positively impact the profession and society, achieving sustainable development.

Course Specification

1- Basic Information

Course Title (according to the bylaw)	Clinical Critical Care and Emergency Nursing			
Course Code (according to the bylaw)	CLN 2424			
Department/s participating in delivery of the course	Critical Care and Emergency Nursing department			
Number of credit hours/points of the course (according to the bylaw)	Theoretical	Practical	Other (specify)	Total
		4		
Course Type				
Academic level at which the course is taught	المستوي / الفرقة الثانى			
Academic Program	Bachelor of Science in Nursing			
Faculty/Institute	Nursing			
University/Academy	AL Ryda for science and technology			
Name of Course Coordinator	Assistant professor /Afaf Mohammed Mansour			
Course Specification Approval Date	1/2/2026			
Course Specification Approval (Attach the decision/minutes of the department /committee/council)				

1.Course Overview (Brief summary of scientific content)

This course focuses on the integration of knowledge and skills to provide safe and effective nursing care to critically ill adults patients with complex needs. Also, emergency nursing care is designed to provide nursing students with skills required to care effectively and safely for seriously injured victims.

2. Course Learning Outcomes CLOs

3- Course specification based on competence:

Domain No. (1) PROFESSIONAL AND ETHICAL PRACTICE

Competency	Key elements	Course Subjects	Course Objectives
1.1.*Demonstrate knowledge, understanding, responsibility and accountability of the legal obligations for ethical nursing practice*	**Demonstrate understanding of the legislative framework and the role of the nurse and its regulatory functions.** **Apply value statements in nurses' code of ethics and professional conduct for ethical decision making.** **Practice nursing in accordance with institutional/national legislations, policies and procedural guidelines considering patient/ client rights.** **Demonstrate responsibility and accountability for care within the scope of professional and practical level of competence**	• **Mechanical ventilation care** • **Tracheostomy tube care**	• **Apply nursing care of MV patients** • **Implement Nursing care to tracheostomy tube patients**

Domain No. (2) HOLISTIC PATIENT-CENTERED CARE

Competency	Key elements	Course Subject	Course Objectives
2.1**Provide holistic and evidence-based nursing care in different practice settings**.	2.1.1 Conduct holistic and focused bio- psychosocial and environmental assessment of health and illness in diverse settings	Advanced cardiac life support	- Apply ethical principles during ACLS &code management -Apply Cardioversion for patients with Defibrillation - Apply management of patients in Cardiopulmonary Arrest - Make clinical decisions during cardiac arrest to save patient life
	2.1.2.Provide holistic nursing care that addresses the needs of individuals, families and communities across the life span	Arterial catheter care Apply arterial line care and sampling	-Perform Arterial line care &sampling -Interpret Arterial line waves Arterial Blood Gases (ABG) -Perform Special consideration when obtain ABG from arterial line

	2.1.6. Examine evidence that underlie clinical nursing practice to offer new insights to nursing care for patients, families, and communities	- pulmonary catheter care	Assist during pulmonary catheter insertion Provide nursing care to pulmonary catheter patients
2.2 Provide health education based on the needs/problems of the patient/client within a nursing framework	2.2.1.Determine health related learning needs of patient/client within the context of culture, values and norms.	- total parenteral nutrition	- Implementation to total parental nutrition for critical ill patients
	2.2.2.Assess factors that influence the patient's and family's ability, including readiness to learn, preferences for learning style, and levels of health literacy	-ICU scoring system -Bundles of care - Common ICU medication	- Apply scoring system for assessment of critical ill patients -Apply bundles of care for critical ill patients - identify common ICU medication and its action and nursing percussion

3. Teaching and Learning Methods

1. **Demonstration**
2. **Redemonstration.**
3. **Case Scenario**
4. **Simulation**
5. **Audiovisual Material (Data show /education video)**

Course Schedule:

Number of the Week	Scientific content of the course (Course Topics)	Total Weekly Hours	Expected number of the Learning Hours			
			Theoretical teaching (lectures/discussion groups/)	Training (Practical/Clinical/)	Self-learning (Tasks/ Assignments/ Projects/ ...)	Other (to be determined)
1	Care of mechanical ventilation	6		1		
2	Tracheostomy tube care	6		1		
3	Advanced cardiac life support	6		1		
4	Arterial catheter care	6		1		
5	Pulmonary catheter care	6		1		
6	Total parental nutrition	6		1		
7	ICU scoring system	6		1		
8	Bundles of care	6		1		
9	Common ICU medication	6		1		

•	Learning Resources and Supportive Facilities *	
Learning resources (books, scientific references, etc.) *	The main (essential) reference for the course (must be written in full according to the scientific documentation method)	• Urden ,L., Stacy, K.,s Lough, M.(2020). Priorities in Critical Care Nursing. 8 th.ed. ELSEVIER Company • Gohnson ,A.,s Crumiett ,H.,(2018). Critical Care Nursing Certification . 7th .ed. New York • Good,V., and Kirkwood P.(2018). Advanced Critical Care Nursing. Elsevier company. 2nd Edition ; 464: 67. • Wiegand ,D., (2017). Procedure Manual For High Acuity, Progressive ,and Critical Care .7th .ed. Elsevier company.
	Other References	
	Electronic Sources (Links must be added)	• Nursing Informatics in Critical CareA systematic review published in BMC Nursing explores how nursing informatics—like electronic health records (EHRs), clinical decision support systems (CDSSs), and telehealth—

		• The American Association of Critical-Care Nurses (AACN) offers several journals and newsletters:
	Learning Platforms (Links must be added)	1. CCRN (Critical Care Registered Nurse) – AACN 2. CEN (Certified Emergency Nurse) – BCE 3. TNCC (Trauma Nursing Core Course) – ENA 4. ACLS s PALS (Advanced Cardiac/Pediatric Life Support) – AHA 5. CCNS (Critical Care Nurse Specialist) – AACN
	Other (to be mentioned)	-
Supportive facilities & equipment for teaching and learning *	Devices/Instruments	Montor, ECG, DC Shock, ambo bag
	Supplies	Syring, cvp manometer
	Electronic Programs	LMS, Microsoft team, Zoom meeting
	Skill Labs/ Simulators	Sem man
	Virtual Labs	Glasses for virtual reality
	Other (to be mentioned)	-

7. Methods of students' assessment

No.	Assessment Methods *	Assessment Timing (Week Number)	Marks/ Scores	Percentage of total course Marks
1	Exam 1written (quiz 1)	4weeks	10	
2	Exam 2 (quiz 2)	8weeks	10	
3	Midterm exam	7week	20	
4	Final Practical/Clinical/… Exam	15 weeks	60	
5	Final Oral Exam	15week	20	
6	Assignments / case study (Activity)	10 weeks	10	
7	Field training (Assessment-care plan-punctuality)		20	
	Total		150	

* The list mentioned is an example, the institution may add and/or delete depending on the nature of the course

Name and Signature Course Coordinator	Name and Signature Program Coordinator

Table of content

Topic	Page
1. Mechanical ventilation	12
2. Tracheostomy Care	30
3.Advanced Cardiac life support	38
4.Arterial Catheter care	56
5.Pulmonary catheter care	63
6.Total Parenteral Nutrition	68
7.ICU scoring system	73
8.Bundles of Care	81
9. Common ICU Medications	84

Mechanical ventilation

Definition

Indications

Mechanical ventilation

Parameters of mechanical ventilation

Alarms of mechanical ventilation

Complications of mechanical ventilation

Equipment

Nursing care

Definition

Mechanical Ventilator is a positive or negative pressure breathing device that can maintain ventilation and oxygen delivery for a prolonged period.

Types of Ventilation

Type	Description
Invasive	**Tube inserted into airway (endotracheal or tracheostomy)**
Non-invasive	**Mask over nose/mouth (BiPAP, CPAP)**

PURPOSES OF MECHANICAL VENTILATION

1. To maintain adequate oxygenation
2. To maintain adequate ventilation
3. To remove carbon dioxide (CO2)
4. To prevent respiratory muscle fatigue
5. To support gas exchange
6. To allow sedation when needed
7. To permit lung healing
8. To maintain airway patency
9. To prevent aspiration

🎯 2. INDICATIONS (When to Use Mechanical Ventilation)

Respiratory Failure

- **Hypoxemic Failure**: Low oxygen (PaO_2 <60 mmHg on room air)
- **Hypercapnic Failure**: High CO_2 ($PaCO_2$ >50 mmHg with acidosis)

Specific Clinical Situations

- Acute respiratory distress syndrome (ARDS)
- Severe pneumonia
- Chronic obstructive pulmonary disease (COPD) exacerbation
- Asthma attack (severe)
- Pulmonary edema
- Acute lung injury

Other Indications

- **Airway Protection**: Decreased level of consciousness (GCS <8)
- **Trauma**: Chest trauma, head injury
- **Post-Operative**: After major surgery requiring sedation
- **Neuromuscular Disorders**: Guillain-Barré syndrome, myasthenia gravis
- **Cardiac Arrest**: Post-resuscitation
- **Shock**: Severe sepsis or cardiogenic shock

. MODES OF MECHANICAL VENTILATION

Volume moods

How It Works:

- Ventilator delivers a SET VOLUME of air with each breath
- Volume is guaranteed - patient gets exact amount set
- Pressure varies depending on lung resistance

Common Volume Modes:

- Volume Control (VC)
- Assist-Control Volume (AC-VC)
- SIMV-Volume

A. Assist-Control (A/C) or CMV (Controlled Mandatory Ventilation)

How it works:

- Ventilator delivers a set number of breaths per minute
- If patient tries to breathe, ventilator assists with full breath
- Every breath (set or patient-triggered) receives full tidal volume

When used:

- Critically ill patients
- Patients who cannot breathe on their own
- Immediate post-intubation

B. SIMV (Synchronized Intermittent Mandatory Ventilation)

How it works:

- Ventilator delivers set number of breaths
- Patients can take additional breaths on their own (without full support)
- Synchronizes with patient's own breathing efforts

When used:

- Weaning from ventilator
- Patients with some breathing ability
- Transitioning to spontaneous breathing

Pressure moods

How It Works:

- Ventilator delivers air at a SET PRESSURE
- Pressure is guaranteed - stays constant
- Volume varies depending on lung compliance

Common Pressure Modes:

- Pressure Control (PC)
- Pressure Support (PS)
- CPAP/BiPAP

PRESSURE CONTROL VENTILATION -

What it is:

- Ventilator delivers breaths at a SET PRESSURE (not volume)
- Pressure remains constant during inspiration
- Volume delivered varies based on lung compliance

When used:

- **1. To prevent high airway pressures** (barotrauma risk)
- **2. To protect injured lungs** (ARDS, pneumonia)
- **3. To improve patient comfort** (better flow pattern)
-

C. Pressure Support (PS)

How it works:

- Patient initiates all breaths
- Ventilator provides pressure support to help
- Patient controls rate and depth

When used:

- Weaning process
- Patients breathing spontaneously but need some help
- Often combined with SIMV

D. CPAP (Continuous Positive Airway Pressure)

How it works:

- Patient breathes spontaneously
- Constant positive pressure throughout breathing cycle
- No ventilator breaths delivered

When used:

- Sleep apnea
- Mild respiratory distress
- Weaning process

D-BiPAP = Bi-level Positive Airway Pressure

A **non-invasive** breathing machine that delivers air through a **mask** (not a tube in the throat)

How It Works

Delivers **TWO levels of pressure:**

Phase	Pressure	Purpose
IPAP (Inspiratory)	**Higher pressure**	Helps you breathe IN
EPAP (Expiratory)	**Lower pressure**	Helps you breathe OUT

BiPAP vs CPAP

	BiPAP	CPAP
Pressure levels	TWO (in and out)	ONE (constant)
Easier for patient	Yes - easier to exhale	No - same pressure throughout
Uses	Respiratory failure, COPD	Sleep apnea mainly

Comparison Table

Feature	Volume Mode	Pressure Mode
What's Fixed	Volume (mL)	Pressure (cmH_2O)
What Varies	Pressure	Volume
Priority	Deliver exact volume	Protect lungs from pressure
Pressure Risk	Can get very high	Limited/controlled
Volume Risk	Guaranteed	Can drop if lungs change
Patient Comfort	Less comfortable	More comfortable
Best For	Sedated/paralyzed patients	Awake/spontaneous breathing

4. VENTILATOR PARAMETERS (Settings)

Parameter	Definition	Normal Range	Rationale
Tidal Volume (VT)	Amount of air delivered per breath	6-8 mL/kg of ideal body weight	Prevents lung injury; lower volumes for ARDS
Respiratory Rate (RR)	Number of breaths per minute	12-20 breaths/min	Maintains adequate ventilation and CO_2 removal
FiO_2	Fraction of inspired oxygen (%)	21%-100% (start 100%, wean to <60%)	Provides adequate oxygenation; high FiO_2 can cause oxygen toxicity
PEEP	Positive pressure at end of expiration	5-20 cm H_2O (usually 5-10)	Prevents alveolar collapse, improves oxygenation
I:E Ratio	Inspiratory to expiratory ratio	1:2 (normal breathing pattern)	Allows adequate time for exhalation
Pressure Support	Extra pressure to assist breathing	5-20 cm H_2O	Reduces work of breathing
Peak Inspiratory Pressure (PIP)	Maximum pressure during breath	<35 cm H_2O	High pressure can cause lung injury
Sensitivity	How easily patient triggers breath	-1 to -2 cm H_2O	Prevents work of breathing but avoids auto-triggering

5. VENTILATOR ALARMS

HIGH PRESSURE ALARM (Most Common)

Causes:

- Patient coughing or biting tube
- Secretions in airway (need suctioning)
- Kinked tubing
- Bronchospasm
- Decreased lung compliance (worsening condition)
- Water in ventilator tubing

Nursing Actions:

1. **Assess patients first!** (ABC - Airway, Breathing, Circulation)
2. Check tube position and patency
3. Suction if needed
4. Check all tubing for kinks
5. Empty water from tubing
6. Auscultate lung sounds
7. Notify provider if condition worsening

LOW PRESSURE ALARM

Causes:

- Disconnection or loose connection
- Cuff leak in ET tube
- Tubing disconnects

Nursing Actions:

1. **Check all connections immediately!**
2. Reconnect any disconnected tubing
3. Check ET tube cuff pressure
4. Manually ventilate with bag-valve-mask if needed
5. Call for help if cannot resolve quickly

LOW TIDAL VOLUME ALARM

Causes:

- Air leak in system
- Cuff leak
- Patient not triggering breaths

Nursing Actions:

1. Check for disconnections
2. Assess cuff pressure
3. Evaluate patient's respiratory effort
4. Check all connections

HIGH RESPIRATORY RATE ALARM

Causes:

- Pain or anxiety
- Hypoxemia
- Fever
- Metabolic acidosis
- Inadequate sedation

Nursing Actions:

1. Assess patient comfort and oxygenation
2. Check vital signs
3. Address pain/anxiety
4. Review ABG results

APNEA ALARM

Causes:

- Over-sedation
- Neurological deterioration
- Ventilator malfunction

Nursing Actions:

1. **Assess patient immediately!**
2. Stimulate patient if over-sedated
3. Manually ventilate if needed
4. Check ventilator settings
5. Call provider immediately

FiO_2 ALARM

Causes:

- Oxygen supply disconnected
- Oxygen tank empty

Nursing Actions:

1. Check Oxygen Source
2. Verify connections
3. Replace oxygen tank if needed

AIRWAY MANAGEMENT (Ongoing)

Nursing Action	Rationale
Verify ET tube position	Prevents accidental extubation or main-stem intubation
Check tube marking at lip/teeth	Ensures tube hasn't migrated
Secure ET tube properly	Prevents dislodgement
Check cuff pressure (20-30 cm H_2O)	Prevents aspiration but avoids tracheal damage
Oral care with chlorhexidine	Reduces VAP risk
Suction only when needed (not routinely)	Remove secretions without causing trauma
Reposition ET tube daily	Prevents pressure ulcers on lips/mouth

EMERGENCY PREPAREDNESS (Always Ready)

Emergency	Immediate Nursing Action
Accidental extubation	Call for help, manually ventilate with bag-valve-mask, prepare for re-intubation
Ventilator malfunction	Disconnect from ventilator, manually ventilate, call for backup ventilator
Cardiac arrest	Disconnect ventilator, begin CPR, manually ventilate between compressions
Severe desaturation	Increase FiO_2 to 100%, suction if needed, assess tube position, call provider
Tension pneumothorax	Notify provider STAT, prepare for chest tube insertion, monitor vital signs

WEANING ASSESSMENT (Daily)

Assessment Criteria	Indicates Readiness for Weaning
Underlying cause improved	**Original problem resolving**
Adequate oxygenation	**SpO_2 >90% on FiO_2 ≤40% and PEEP ≤5-8**
Hemodynamically stable	**No vasopressors or low doses**
Able to protect airway	**Gag and cough reflex present**
Alert and cooperative	**Follows commands**
Adequate respiratory effort	**Negative inspiratory force >-20 cm H_2O**
Successful spontaneous breathing trial	**Tolerates 30-120 minutes on minimal support**

Nursing Role in Weaning:

- Perform daily assessments
- Conduct spontaneous breathing trials as ordered
- Monitor closely during trials
- Report readiness to provider
- Encourage and support patient

SYSTEM	FAILURE SIGNS	Numbers
BREATHING	• Breathing too fast • Breathing too slow • Small, shallow breaths • Using neck/chest muscles • Gasping for air	• RR >35 or <8 • TV <5 mL/kg • **RSBI >105**

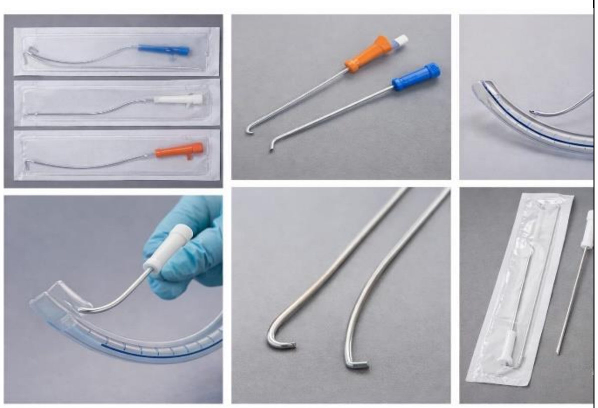
SYSTEM	FAILURE SIGNS	Numbers
OXYGEN	• Low oxygen levels • Lips/skin turning blue	• SpO_2 <90% • PaO_2 <60 mmHg
HEART	• Heart racing • BP too high or too low • Irregular heartbeat	• HR >140 • BP >180 or <90 • New arrhythmias
BRAIN	• Confused/drowsy • Very anxious/agitated • Sweating heavily • Looks scared/panicked	• Decreased alertness • Can't follow commands • Diaphoresis

Weaning Failure - STOP WEANING If You See These:

Complications of mechanical ventilation and nursing prevention:

Complication	Prevention Strategy	Rationale
VAP	HOB elevation, oral care, daily assessment for extubation	Most common ventilator complication
Pressure ulcers	Turn every 2 hours, use pressure-relief devices	Immobility increases risk
DVT/PE	Sequential compression devices, anticoagulation therapy	Immobility and critical illness increase clot risk
Stress ulcers	Administer prophylaxis (H2 blockers, PPIs)	Critical illness increases gastric acid
Muscle weakness	Early mobility, physical therapy when stable	Prolonged immobility causes deconditioning
Barotrauma	Monitor airway pressures, use lung-protective strategies	High pressures can rupture alveoli

Equipment

No.	Equipment	Used For/Purpose
1.	Mechanical Ventilator	Main device that delivers controlled breaths to patients; regulates volume, pressure, rate, and oxygen concentration
2.	Endotracheal Tube (ETT)	Inserted through mouth/nose into trachea to establish airway and connect patient to ventilator
3.	Tracheostomy Tube	Inserted through surgical opening in neck for long-term ventilation; provides airway access
4.	Laryngoscope + Blades	Used to visualize vocal cords during endotracheal intubation
5.	Stylet 	Rigid guide inserted into ETT to maintain shape and facilitate intubation
6.	Water Trap	Collects condensation from humidified gases in circuit
7.	Bacterial/Viral Filter	Prevents contamination between patient and ventilator; filters microorganisms
8.	Oxygen Supply (Wall Outlet/Cylinder)	Provides oxygen source for ventilator to deliver to patient
9.	Medical Air Supply	Provides compressed air source for mixing with oxygen
10.	Oxygen Flowmeter	Regulates and measures oxygen flow rate

No.	Equipment	Used For/Purpose
11.	Heated Humidifier 	Warms and humidifies inspired gases to prevent airway drying and maintain normal respiratory function
12.	Sterile Water	Used to fill humidifier chamber
13.	Pulse Oximeter + Probe	Continuously monitors oxygen saturation (SpO2) in blood
14.	Manual Resuscitation Bag (Ambu Bag)	Provides manual ventilation during transport, emergencies, or ventilator failure
15.	Syringe (10-20 ml)	Used to inflate/deflate cuff on artificial airway

ETT care Procedure:

Step	Procedure	Rationale
1	Hand Washing	Prevents transmission of microorganisms and reduces risk of infection
2	Gather Equipment (suction catheter, sterile gloves, sterile saline, oral care supplies, tape/holder, stethoscope, Ambu bag)	Ensures all necessary items are available; saves time and maintains efficiency
3	Explain Procedure to Patient	Reduces anxiety; promotes cooperation; respects patient autonomy
4	Position Patient (semi-Fowler's 30-45°)	Facilitates drainage of secretions; reduces aspiration risk; improves oxygenation

5	Perform Hand Hygiene & Wear PPE (gloves, mask, goggles)	Protects nurse from bodily fluids; maintains infection control standards
6	Assess Patient (vital signs, SpO2, breath sounds, respiratory effort)	Establishes baseline; identifies any respiratory distress before procedure
7	Pre-oxygenate Patient (increase FiO2 by 10-20% or 100% for 30-60 seconds)	Prevents hypoxemia during suctioning; maintains adequate oxygen levels
8	Perform Endotracheal Suctioning (if needed - insert catheter without suction, apply suction while withdrawing, limit to 10-15 seconds)	Removes secretions obstructing airway; maintains patent airway; prevents hypoxia
9	Assess Breath Sounds (auscultate bilaterally)	Confirms proper tube placement; detects tube migration or obstruction
10	Check ETT Position (verify cm marking at lip/teeth)	Ensures tube hasn't moved; prevents accidental extubation or right mainstem intubation
11	Check Cuff Pressure (maintain 20-30 cmH2O)	Prevents tracheal damage from over-inflation; prevents aspiration from under-inflation
12	Secure ETT (replace tape/holder if soiled; ensure firm but not tight)	Prevents tube displacement; avoids pressure ulcers; maintains airway security
13	Provide Oral Care (clean mouth, teeth, tongue with oral swabs/toothbrush)	Reduces bacterial colonization; prevents ventilator-associated pneumonia (VAP); promotes comfort
14	Inspect Skin (check corners of mouth, lips, nares for pressure areas)	Detects early signs of pressure injury; allows repositioning of tube to prevent skin breakdown
15	Reposition ETT (move tube to opposite side of mouth if appropriate)	Prevents pressure ulcers at lip/mouth corners; promotes skin integrity
16	Reassess Patient (vital signs, SpO2, breathing pattern, comfort level)	Evaluates patient tolerance of procedure; detects complications early
17	Return FiO2 to Baseline (if increased for suctioning)	Prevents oxygen toxicity; maintains appropriate oxygenation levels
18	Ensure Patient Comfort (reposition patient, provide communication method)	Promotes patient well-being; reduces anxiety; maintains dignity
19	Dispose of Equipment & Remove PPE	Prevents cross-contamination; follows infection control protocols
20	Perform Hand Hygiene	Final barrier against infection transmission; completes infection

		control cycle
21	Documentation (date/time, ETT position/cm marking, cuff pressure, secretion characteristics, breath sounds, patient tolerance, complications)	Provides legal record; ensures continuity of care; tracks changes in patient condition; communicates with healthcare team

Routine nursing care

Nursing Action	Rationale
Auscultate lung sounds bilaterally	Assess air entry and detect complications (pneumothorax, atelectasis)
Monitor vital signs (HR, BP, RR, SpO_2)	Detect hemodynamic changes and adequacy of oxygenation
Assess level of consciousness	Determines sedation needs and neurological status
Check chest rise and fall symmetry	Ensures proper ventilation; asymmetry suggests pneumothorax or tube malposition
Monitor oxygen saturation continuously	Early detection of hypoxemia
Assess skin color and capillary refill	Indicates perfusion and oxygenation status
Check ventilator settings match orders	Ensures patients receive correct treatment
Review ABGs (Arterial Blood Gases)	Evaluates effectiveness of ventilation and oxygenation

Care of the mechanical ventilator:

Daily Checks:

1. check alarm settings (high/low pressure, disconnect, apnea)
2. verify ventilator parameters (rate, tidal volume, FiO2, PEEP)
3. inspect breathing circuit (leaks, kinks, water accumulation)
4. empty water traps (prevent obstruction)
5. check humidification system (water level, temperature)

6- clean external surfaces **(prevent infection)**

7- check tubing connections **(tight and secure)**
8- To monitor ventilator graphics **(pressure, volume, flow waveforms)**

Circuit Care:

change circuit every 7 days **(or per protocol/if visibly soiled)**

To replace filters regularly **(bacterial/viral filters)**

To change humidifier water daily **(use sterile water only)**

Tracheostomy care

Outlines: -

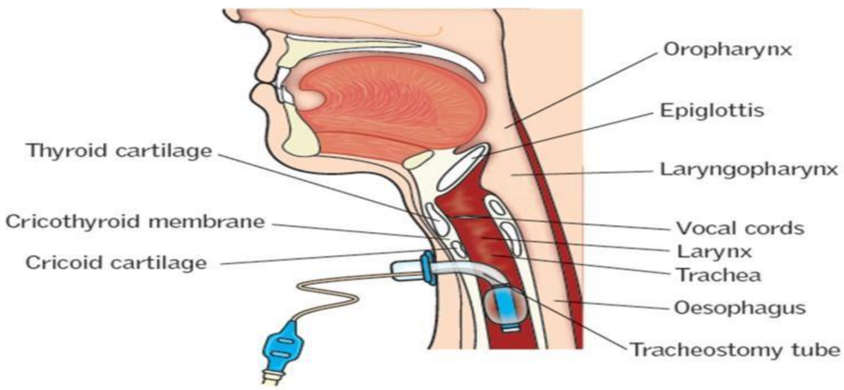
- Definition of the tracheostomy.
- Purpose of the tracheostomy.
- Indications of tracheostomy.
- Types of tracheostomies.
- Describe the tracheostomy tube
- Complications of the tracheostomy.
- Equipment
- Procedure.

Definition

A tracheostomy is a surgical opening through the neck into the trachea. A tracheostomy tube is inserted at the time of surgery to maintain open airway.

The aim of tracheostomy is to bypass obstructions in the upper airway; to aid assisted ventilation.

Caring for a patient with tracheostomy requires the nurse to have a thorough understanding of airway management. Critical situations would require immediate intervention to ensure that respiratory arrest is avoided.



Tracheostomy surgical procedure:

Surgical procedure is done in the operating room or at the bedside, under general anesthesia, and a skin incision (stoma) is created below the cricoid cartilage, then tracheostomy

Types of tracheostomy:

1. **Temporary tracheostomy;** an elective procedure e.g. at the time of major head and neck surgery.
2. **Permanent tracheostomy;** total laryngectomy.

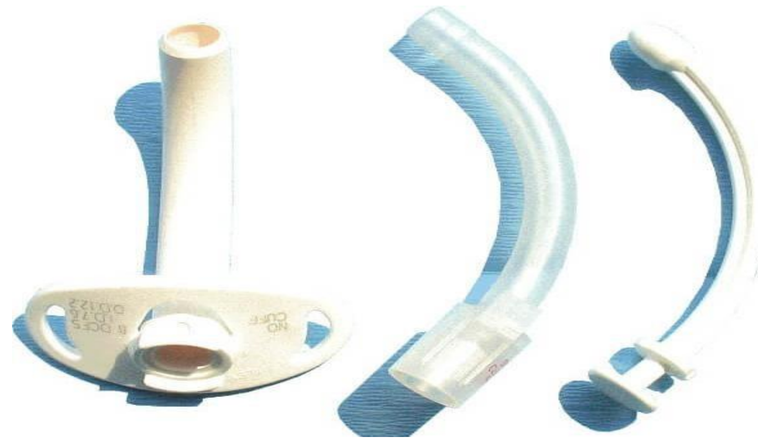
Indications of tracheostomy:

1. Upper airway obstruction caused by trauma, foreign body, tumor (cancerous larynx).
2. Difficulty clearing secretions (copious secretions) from the airway.
3. Prolonged mechanical ventilation (more than 14 days) to avoid vocal cord dystrophy.
4. Inability to endotracheal intubation.

Types of tracheostomy tube:

1. **Cuffed Tube;** Used to obtain a closed circuit for mechanical ventilation.

2. **Cuff less Tube:** Used for patients who are ready for decannulation, may be single or double tube.



Non cuffed tube

Cuffed tube

Complications of Tracheostomy:

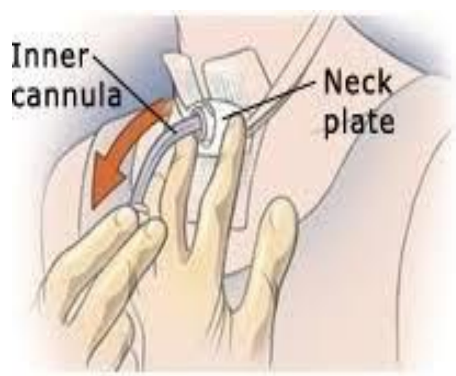
- Tube Dislodgement and Loss of Airway
- Skin Breakdown
- Tracheoesophageal Fistula
- Infection
- Hemorrhage

Airway obstruction

Equipment.

- Sterile gloves
- Clean gloves
- Sterile tracheostomy dressing
- Sterile gauze
- Normal saline
- Hydrogen peroxide (if policy allows)
- Suction catheter and suction apparatus
- Sterile cotton-tipped applicators
- New tracheostomy ties or Velcro holder
- Scissors
- Oxygen source and pulse oximeter

Procedure of Tracheostomy tube care:

Steps	Rational
1. Wash Hands	Reduce transmission of microorganisms
2. Wearing PPE	Save time and effort
3.Hyperoxygenate and suction the trachea and pharynx as needed	Reduces the risk of hypoxemia and arrhythmias; removes secretions and diminishes the patient's need to cough during the procedure.
4. Remove and discard soiled tracheostomy dressing and remove gloves	Provides access and visibility of the tracheostomy site.
5. Wash Hands and wear another gloves	Reduce transmission of microorganisms
6. *For disposable inner cannulas: A. Open prepackaged inner cannula. B. Remove soiled inner cannula. C. Replace with prepackaged cannula. *. **For non-disposable inner cannula** (open prepackaged commercial tracheostomy care kit): A. Prepare sterile saline or sterile water on a sterile field.	Replaces the inner cannula. 
B. Apply sterile gloves. C. Remove oxygen source and then inner cannula, place it in sterile saline or sterile water.	

8. Remove inner cannula against clock wise	To facilitate removal
9-Place tracheostomy collar, T tube, or ventilator oxygen source over or near the outer cannula.	To maintain oxygen supply.
10. Clean the inner cannula with a small brush.	Assists in the removal of debris and thick secretions.
11.Rinse inner cannula by pouring normal saline over the cannula.	To remove any remaining debris from the cannula
12-Dry inner cannula by pipe cleaner	To avoid infection
13. Insert the inner cannula and lock it into place on clock wise.	To maintain proper insertion
14-Clean the skin around the stoma by gently patting in circular motion from far side to near side.	To avoid cross contamination
15. Dry the skin around the stoma by gently patting it dry.	Decreases the likelihood of microorganism growth and skin breakdown.
Apply lubricant on the skin around the stoma	To prevent skin dryness or injury
16.Place a gauze pad V shape between the stoma and tracheostomy.	
17. Change tracheostomy tie by inserting new tie from one side then keep it around the neck to opening other side then make the knot.	To prevent contamination
18- Put two fingers under new tie .to ensure that the tie not too tight and not too loose	To avoid pressure on carotid artery
19. Apply hyperoxygenation and suction the trachea and pharynx.	
20 . Reassess the patient's respiratory status (O2 level, R.R, and depth)	

21. Discard used supplies and remove personal equipment	
22. Wash hands	
23. Document the procedure	

Advanced Cardiac Life Support (ACLS)

Outline:

- Definition
- Indications
- ACLS Medications
- Adult Cardiac Arrest Algorithm
- Bradycardia with Pulse Algorithm
- Tachycardia with Pulse Algorithm
- Post–Cardiac Arrest Care Algorithm
- Resuscitation Team Dynamics
- ACLS Checklists

Purpose

To provide prompt, evidence-based, advanced interventions in adult patients experiencing **cardiac arrest** or other life-threatening cardiac events, aiming to restore **spontaneous circulation** and minimize neurological injury.

Definition

Advanced Cardiovascular Life Support (ACLS) is a set of **evidence-based clinical protocols** designed for the management of adult patients experiencing **cardiac arrest, life-threatening arrhythmias, or other cardiopulmonary emergencies**. ACLS integrates high-quality cardiopulmonary resuscitation (CPR) with advanced airway management, cardiac rhythm recognition, defibrillation, pharmacologic therapy, and post–cardiac arrest care.

Indications

1 Cardiac Arrest

Initiate ACLS when a patient is **unresponsive, not breathing normally, and pulseless**, including:

- **Ventricular fibrillation (VF)**
- **Pulseless ventricular tachycardia (pVT)**
- **Asystole**
- **Pulseless electrical activity (PEA)**

➞ These rhythms require high-quality CPR, electrical therapy (defibrillation or cardioversion), advanced airway management, IV/IO access, and resuscitation drugs.

2 Hemodynamically Unstable Arrhythmias

ACLS is indicated *before full arrest* when arrhythmias are causing poor perfusion, specifically:

- **Symptomatic bradycardia** (with hypotension, shock, altered mental status, ischemia, or acute heart failure)
- **Tachyarrhythmias with a pulse** that cause instability (hypotension, acute altered mental status, signs of shock, ischemic chest pain, acute heart failure)

➞ Immediate rhythm control with pacing, anti-arrhythmic drugs, or synchronized cardioversion may be required.

3 Acute Coronary Syndromes (ACS)

ACLS is indicated for patients with:

- **Suspected or confirmed ST-elevation myocardial infarction (STEMI)**
- **High-risk non-STEMI** features
  These patients require continuous monitoring, early recognition of life-threatening arrhythmias, and prompt advanced intervention.

4 Respiratory Arrest or Failure

When a patient is **apneic or has inadequate ventilation/oxygenation** that cannot be managed with basic airway support alone:

- Advanced airway (endotracheal intubation or supraglottic device)
- Mechanical ventilation
- Oxygenation and capnography monitoring

➞ ACLS is initiated to prevent cardiopulmonary arrest and restore effective breathing.

5 Shock States with Cardiovascular Instability

Patients with shock (cardiogenic, distributive, hypovolemic) and:

- **Systolic BP < 90 mmHg**, signs of organ hypoperfusion, or ongoing deterioration

- ➞ Advanced hemodynamic management, fluids, vasopressors, inotropes, and monitoring may be required

Drug	Use / Rhythm	Dose (Adult)	Key Action
Epinephrine	Cardiac arrest (VF, pVT, PEA, Asystole)	1 mg IV/IO every 3–5 min	↑ BP & perfusion to heart/brain
Amiodarone	Refractory VF/pVT, VT with pulse	300 mg IV/IO (2nd dose 150 mg)	Stabilizes heart rhythm
Lidocaine	Alternative for VF/pVT	1–1.5 mg/kg IV/IO	Suppresses abnormal heart beats
Magnesium	Torsades de pointes	1–2 g IV	Protects heart cells, stops arrhythmia
Atropine	Symptomatic bradycardia	1 mg IV every 3–5 min (max 3 mg)	↑ Heart rate by blocking vagus nerve
Dopamine	Bradycardia if atropine fails	5–20 mcg/kg/min IV	↑ Heart rate & BP
Epinephrine infusion	Bradycardia if atropine fails	2–10 mcg/min IV	↑ Heart rate & BP

Drug	Use / Rhythm	Dose (Adult)	Key Action
Adenosine	SVT (fast regular narrow QRS)	6 mg rapid IV → 12 mg if needed	Stops AV nodal re-entry
Diltiazem	AF/flutter with fast rate	0.25 mg/kg IV	Slows heart rate
Beta-blocker	AF/flutter with fast rate	5 mg IV	Slows heart rate
Procainamide	VT with pulse	20–50 mg/min IV	Stabilizes rhythm
Sotalol	VT with pulse	100 mg IV over 5 min	Stabilizes rhythm

List of equipment

1. Airway and Ventilation Equipment

- Bag-valve-mask (BVM) with oxygen reservoir
- Oxygen supply with regulator and tubing
- Suction device (manual or electric) with catheters
- Oropharyngeal airways (OPA) of various sizes
- Nasopharyngeal airways (NPA)
- Endotracheal tubes (ETT) of various sizes
- Laryngoscope with blades (Macintosh and Miller)
- Video laryngoscope (optional but useful)
- Stylet or bougie for difficult intubation
- Capnography or end-tidal CO_2 monitor (for confirming ETT placement)
- Mechanical ventilator (if available in hospital settings)

2. Circulation and Monitoring Equipment

- Cardiac monitor / defibrillator (with pacing capabilities)
- Electrocardiogram (ECG) leads and electrodes
- Defibrillator pads (adult and pediatric sizes)
- External pacemaker (transcutaneous)
- Blood pressure monitor (manual or automated)
- Pulse oximeter

3. Vascular Access and IV/IO Equipment

- IV cannulas (various sizes)
- Intraosseous (IO) needles and drill (for emergency vascular access)
- Normal saline or Ringer's lactate fluids
- IV tubing and extension sets
- Syringes and needles for medication administration
- Saline flushes

4. Medications Commonly Used in ACLS

(While technically not "equipment," they are essential for ACLS protocols)

- Epinephrine
- Amiodarone
- Adenosine
- Atropine
- Magnesium sulfate (for torsades de pointes)
- Sodium bicarbonate (for certain cases)
- Vasopressin (if included in protocol)

5. Miscellaneous and Supportive Equipment

- Thermometer
- Glucometer (to check blood glucose in altered mental status)
- Emergency crash cart (containing all essential drugs and equipment)
- Personal protective equipment (PPE: gloves, masks, eye protection)
- Scissors or trauma shears
- Clock or timer (to track compressions, defibrillation, and drug administration)
- Documentation forms / electronic medical record access

1) Adult Cardiac Arrest Algorithm (ACLS)

(For unresponsive adult with **no pulse** and not breathing normally)

Initial actions

1. **Activate emergency response system** and **start high-quality CPR** immediately.
2. **Attach monitor/defibrillator**; assess rhythm.
3. Ensure appropriate airway and oxygenation.

Shockable rhythm — VF / pulseless VT

- Deliver shock (defibrillation) immediately.
- Resume CPR for 2 minutes (minimize interruptions).
- Establish IV/IO access.
- **Epinephrine** every 3–5 min.
- For refractory VF/pVT: **Amiodarone or Lidocaine** per dosing.
- Continue alternating CPR, rhythm checks, and shocks until ROSC or termination.

Non-shockable rhythm — Asystole / PEA

- Resume CPR for 2 minutes.
- Administer **epinephrine** every 3–5 minutes.
- Continue high-quality CPR and rhythm checks.
- Treat **reversible causes (H's and T's).**

When ROSC (return of spontaneous circulation) achieved → proceed to **Post–Cardiac Arrest Care** immediately.

2) Adult Bradycardia with a Pulse Algorithm

Assessment

- **Bradycardia** often defined as heart rate typically <50/min in symptomatic patients.
- Evaluate for **cardiopulmonary compromise:** hypotension, altered mental status, shock, ischemic chest discomfort, acute heart failure.

Management steps

1. **Maintain airway and oxygen; monitor cardiac rhythm.**
2. If **no signs of compromise:** observe and identify underlying causes.
3. If **cardiopulmonary compromise present:**
   - **Atropine 1 mg IV** (repeat every 3–5 min up to 3 mg).
4. If atropine **ineffective:**
   - Consider **transcutaneous pacing**
   - **Dopamine infusion (5–20 mcg/kg/min)** or **epinephrine infusion (2–10 mcg/min)**.
5. Seek **expert consultation**; consider transvenous pacing if needed.

3) Adult Tachycardia with a Pulse Algorithm

Assess stability

- **Unstable tachycardia** signs: hypotension, altered mental status, shock, ischemic chest discomfort,
- acute heart failure — proceed with urgent intervention.
- **Stable tachycardia** may be managed medically.

If unstable

- **Synchronized cardioversion** immediately (sedate if possible).
- Follow with appropriate energy settings based on rhythm type.

If stable

1. Evaluate **QRS width**:
   - **Narrow-complex tachycardia (<0.12 s):** vagal maneuvers; **adenosine** if regular; consider β-blockers
   - or calcium channel blockers.
   - **Wide-complex tachycardia (≥0.12 s):** treat as VT if monomorphic; use antiarrhythmics like **amiodarone**; consult expert if uncertain.
2. Address underlying causes (e.g., electrolyte imbalance, ischemia).

4) Adult Post–Cardiac Arrest Care Algorithm

(Begins **after ROSC**)

Initial stabilization

1. **Airway and breathing**
   - Assess airway; consider advanced airway.
   - Titrate oxygen to maintain **SpO_2 90–98%**.
   - Target **$PaCO_2$ 35–45 mm Hg**.
2. **Hemodynamics**
   - Maintain **mean arterial pressure (MAP) ≥ 65 mm Hg**.
   - Fluids and vasopressors as needed.
3. **Diagnostic testing**
   - **12-lead ECG** to identify ischemia/arrhythmias.
   - Consider imaging (CT/ultrasound) if indicated.
4. **Identify and treat causes**
   - Coronary angiography for STEMI or high-risk features.
   - Treat arrest etiology and complications.
5. **Temperature control**
   - Consider **targeted temperature management** for comatose patients (goal ~32–37.5 °C).

- **Core temperature** is the internal body temperature of the deep organs and central circulation that reflects true physiological heat content and is used to guide clinical management in critical care settings.

6. **Ongoing care**
   - EEG evaluation for seizures, multimodal neurologic prognostication, ICU care.

Resuscitation Team Dynamics

High-performance team coordination improves resuscitation outcomes.

1. **Team leader** assigns roles and directs the resuscitation effort.
2. **Team leader** monitors performance and corrects deficiencies.
3. **Team leader** remains at the foot of the bed for optimal oversight.
4. **Airway provider** manages airway patency, oxygenation, and ventilation.
5. **Medication provider** establishes IV/IO access and administers medications and fluids.
6. **Monitor/defibrillator operator** applies pads, displays rhythm, and delivers shocks as ordered.
7. **Compressor** provides high-quality chest compressions with minimal interruptions.
8. **Compressor** rotates every 2 minutes or 5 cycles to prevent fatigue.
9. **Recorder** documents events, medications, and timing during resuscitation.
10. **Recorder** prompts time-sensitive interventions such as epinephrine dosing.
11. **All team members** use closed-loop communication to confirm and complete orders.
12. **All team members** maintain mutual respect and clear communication

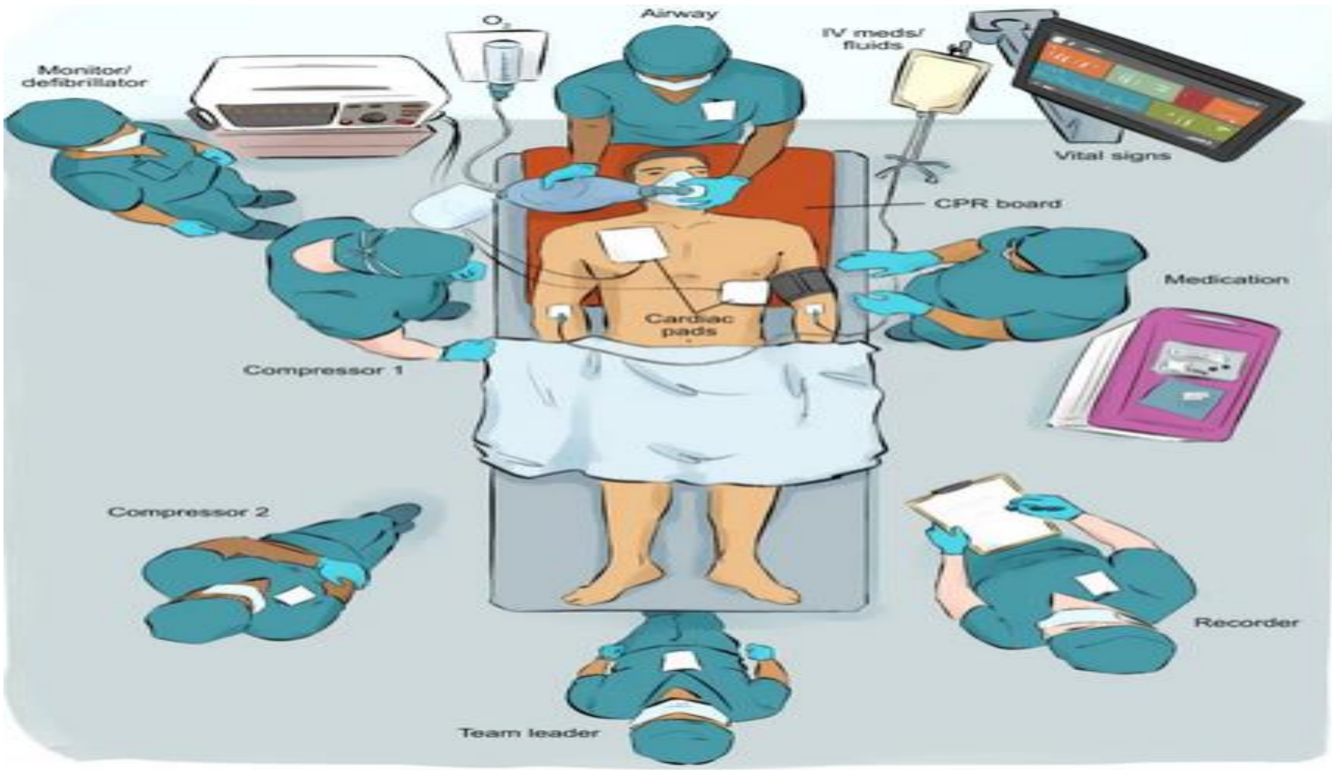
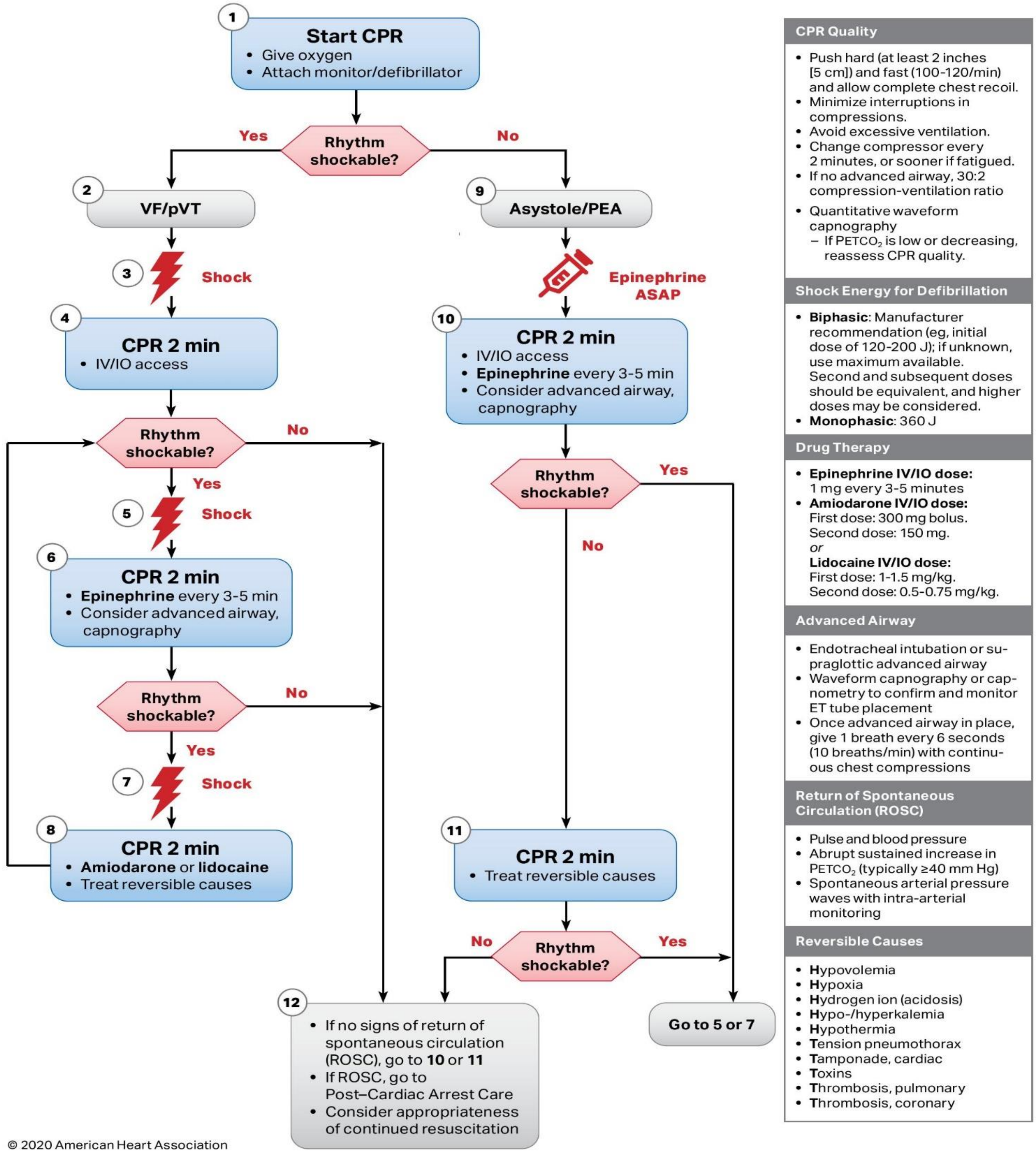
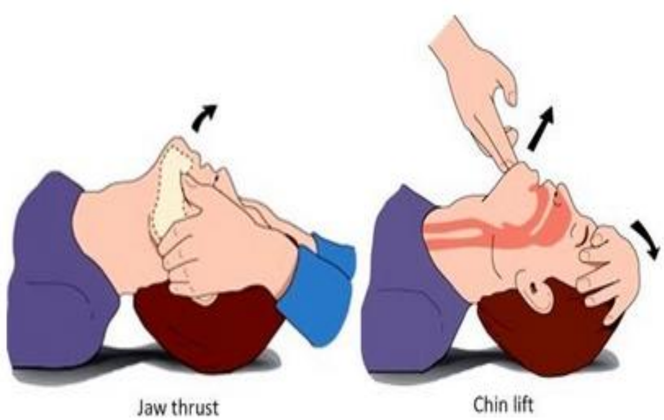


Figure 19 - An example of optimized team placement during resuscitation

Adult Cardiac Arrest Algorithm



ACLS procedure checklist (step with rationale)

Step	Action	**Rationale**
	1. Immediate Recognition and Activation	
1.	Ensure scene safety .	To protect the rescuer(s) and healthcare team from harm.
2.	Assess unresponsiveness: Tap and shout	Confirms patient status and urgency
3..	Call for help, activate emergency response team/code blue	Ensures immediate team mobilization
4..	Simultaneously check for breathing and carotid pulse (max 10 seconds)	Determines need for CPR and whether cardiac arrest is present
5.	If no pulse, initiate high-quality CPR immediately	Early CPR improves perfusion and survival
	2. Start High-Quality CPR	
6.	Begin chest compressions: depth 5–6 cm, rate 100–120/min, full recoil 	Maximizes cardiac output during arrest
7.	Use 30:2 compression-to-ventilation ratio if no advanced airway	Ensures adequate oxygenation while maintaining perfusion
	Open airway using jaw-thrust or head-tilt-chin-lift	Ensures airway patency
8.	Provide bag-valve-mask ventilation with 100%  oxygen	Supports oxygenation during arrest
9.	**Attach** defibrillator/monitor as soon as available	Enables rhythm identification for appropriate treatment

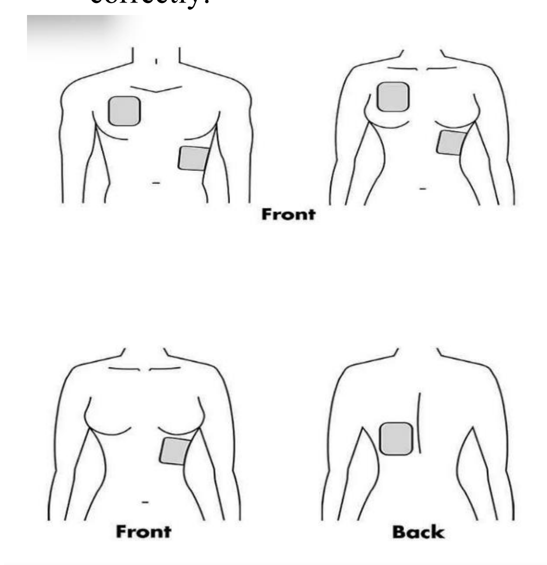
<table>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td>3.Rhythm Analysis and Defibrillation</td><td></td></tr>
<tr><td>10.</td><td>Analyze rhythm (VF/pVT vs PEA/asystole)

Rhythm analysis

○ Briefly pause CPR (≤10 seconds).
○ Identify the rhythm on the monitor.
○ Determine shockable (VF/pVT) vs non-shockable (PEA/asystole).</td><td>Determines whether rhythm is shockable</td></tr>
<tr><td>11.</td><td>⬇ Only if the rhythm is shockable ⬇

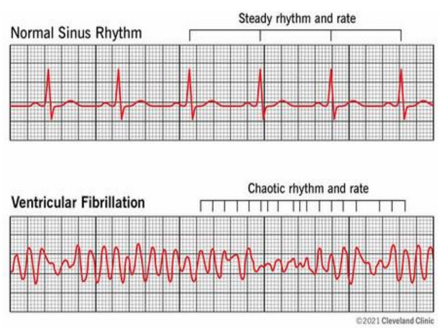
Shockable rhythm (VF/pVT): Deliver 1 shock (200 J biphasic),

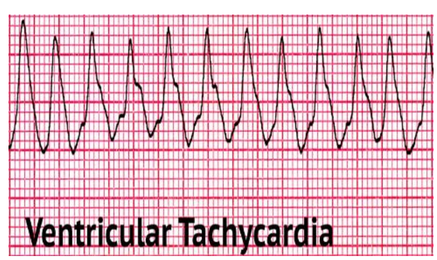
Prepare for shock delivery

• Ensure defibrillator pads are attached correctly.


• Charge the defibrillator.
• Continue CPR while charging (if possible)

Prepare for Shock After rhythm analysis
(Select the appropriate energy level):

○ Biphasic: typically, 120–200 J (per manufacturer)</td><td>Early defibrillation improves survival




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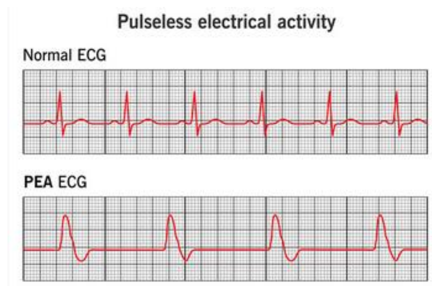
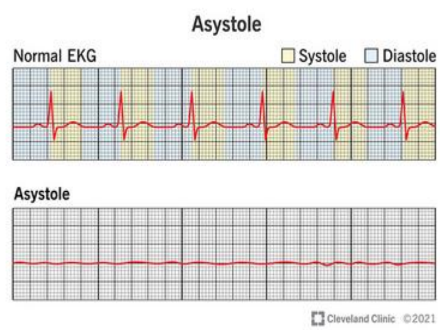
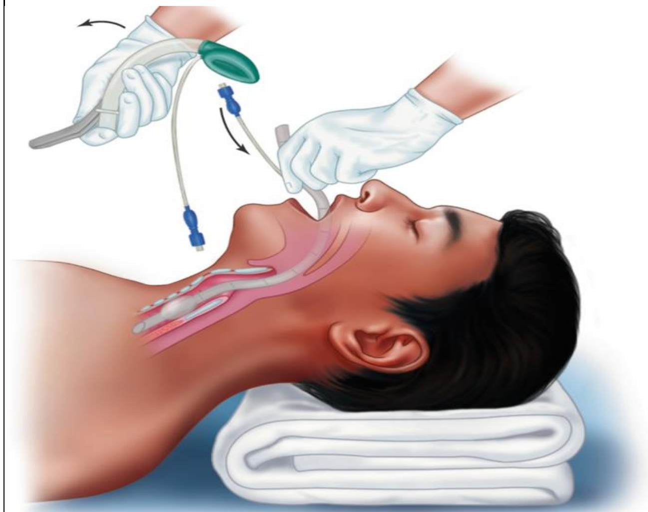
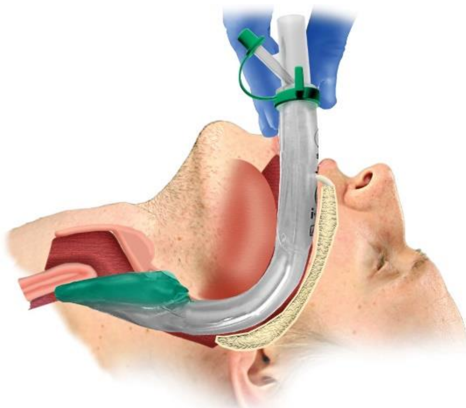
</table>

	◦ Monophasic: 360 J ◦ Charge the defibrillator while minimizing interruptions in CPR ◦ keep all precautions of using defibrillator as mentioned **safety check** • Say “CLEAR!” before shocking; ensure no contact • Move oxygen source away from patient’s chest. **Then** **Deliver shock promptly.** • resume CPR immediately for 2 minutes	
12.	**If** Non-shockable rhythm (PEA/asystole): Continue CPR, do not shock	 These  rhythms don’t benefit from defibrillation
13.	Reassess rhythm every 2 minutes	Guides further intervention based on rhythm evolution
	4. Airway and Breathing	
	• Open **the Airway** • Use **jaw-thrust** if spinal injury suspected, or **head-tilt-chin-lift** otherwise. • Provide **Bag-Valve-Mask (BVM) Ventilation** • Use **100% oxygen**. • Rate: **2 breaths after every 30 compressions** (if no advanced airway). • Place **Advanced Airway (ETT or Supraglottic**	• Rationale: Ensures **airway patency** to allow effective ventilation. • Rationale: Supports **oxygen delivery** during resuscitation.

	Device like LMA (Laryngeal Mask Airway)   • Only by trained provider.	• Rationale: Allows **continuous chest compressions without pauses**, ensures **definitive ventilation**, and provides **capnography monitoring**. • • Rationale: Prevents **hyperventilation**, maintains **oxygenation and perfusion**, improves **ROSC chances**. • Rationale: $ETCO_2$ is a **marker of CPR quality and circulation**

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- Use **$ETCO_2$ (waveform capnography)** to confirm placement.

• Ventilation **After Advanced Airway Placement**

- **Rate:** 1 breath every 6 seconds (10/min)
- **Compressors:** Continue uninterrupted CPR

- **Monitor $ETCO_2$**

- Target: **>10 mmHg during CPR**
- Sudden rise may indicate **ROSC**

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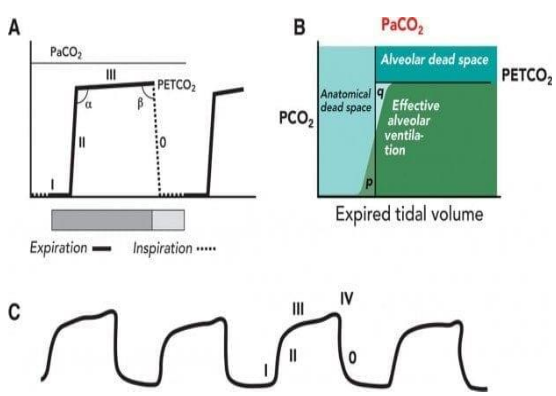

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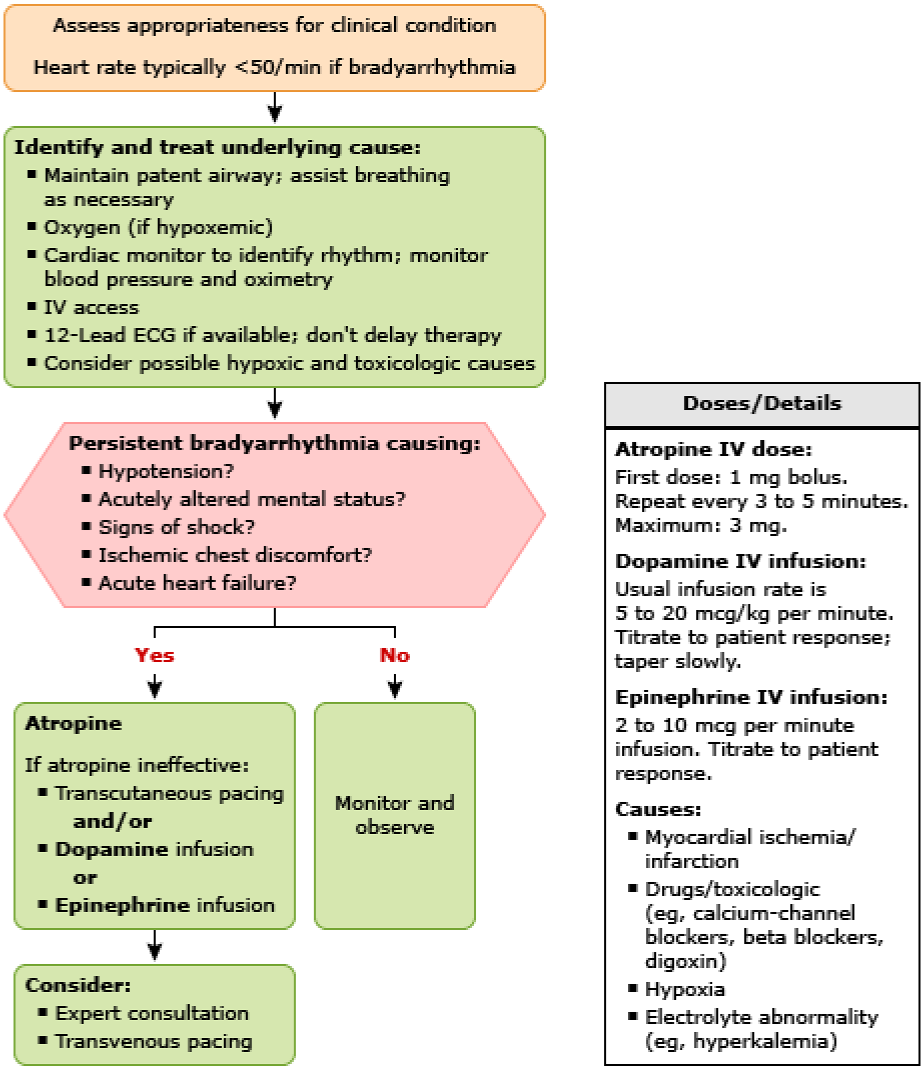
<tr><td></td><td>5. IV/IO Access and Medication Administration</td><td></td></tr>
<tr><td>19.</td><td>Establish IV or IO access</td><td>Enables rapid medication delivery</td></tr>
<tr><td>20.</td><td>Epinephrine 1 mg IV/IO every 3–5 minutes</td><td>Increases coronary and cerebral perfusion pressure</td></tr>
<tr><td>21.</td><td>For VF/pVT: Give Amiodarone 300 mg IV push (2nd dose: 150 mg) or Lidocaine alternative</td><td>Treats refractory shockable rhythms</td></tr>
<tr><td>22</td><td>Flush line after each drug with 20 mL saline</td><td>Ensures rapid delivery into circulation</td></tr>
<tr><td>23.</td><td>6.Identify and Treat Reversible Causes (Hs & Ts)</td><td></td></tr>
<tr><td></td><td>1. Hypoxia</td><td>• Ensure airway/oxygen delivery</td></tr>
</table>

	2. Hypovolemia 3. Hydrogen ion (acidosis) 4. Hypo-/Hyperkalemia 5. Hypothermia 6. Tension pneumothorax 7. Tamponade (cardiac) 8. Toxins 9. Thrombosis (coronary) 10. Thrombosis (pulmonary)	• Administer IV fluids, blood • Sodium bicarbonate if indicated • Calcium chloride, insulin-glucose • Warm the patient • Needle decompression • Pericardiocentesis • Antidotes (e.g., naloxone) • PCI / thrombolysis • Thrombolytics
	7. Return of Spontaneous Circulation (ROSC)	
24.	Check for pulse and spontaneous breathing	Confirms ROSC
25.	Optimize oxygenation: Maintain SpO_2 94–99%	Prevents both hypoxia and hyperoxia
26.	Monitor BP; consider fluids or vasopressors	Supports hemodynamics
27.	Obtain 12-lead ECG	Identifies myocardial infarction or arrhythmias
28.	**Consider Targeted Temperature Management** • Consider TTM in unresponsive patients • Maintain **core** temperature 32–36 °C for 24 hours **or** • Maintain normothermia (36–37.5 °C) with strict prevention of **hyperthermia**	Preserves neurological function
29.	Transfer to ICU for further post-arrest care	Requires continuous monitoring and support
	8. Documentation and Communication	
30.	Document rhythm changes, medications, times, response to interventions	Ensures accurate records for evaluation/legal purposes
31.	Communicate clearly using closed-loop communication	Reduces errors in team-based resuscitation
32	Debrief with team post-event	Improves future performance and identifies gaps



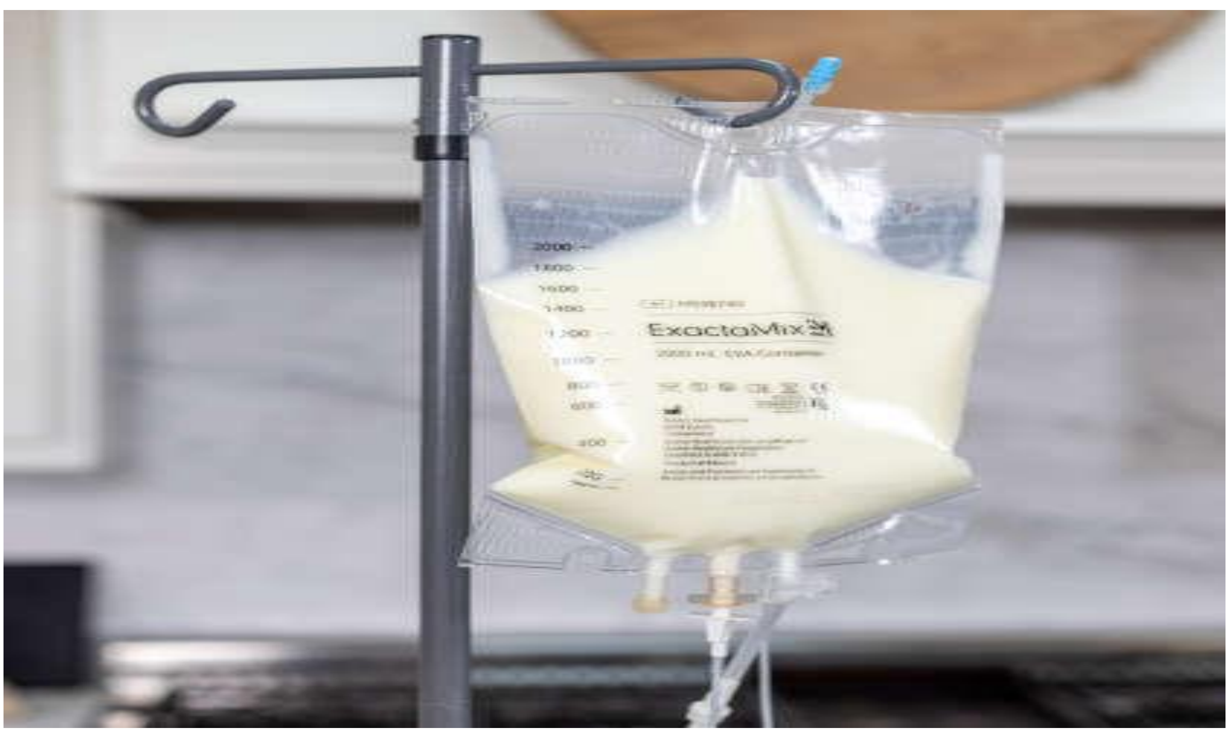
ADULT BRADYCARDIA WITH A PULSE ALGIORHYSM

ACLS CHECKLIST — BRADYCARDIA WITH A PULSE (AHA 2025)

Step	Action	Rationale
1	Ensure scene and provider safety	Protects healthcare team
2	Assess ABCs — airway, breathing, circulation	Initial stabilization
3	Attach cardiac monitor, pulse oximetry, BP cuff	Continuous rhythm and hemodynamic monitoring
4	Establish IV access	Enables drug administration
5	Assess for cardiopulmonary compromise (hypotension, shock, altered mental status, ischemia, acute HF)	Determines need for immediate treatment
6	Provide oxygen if hypoxemic	Improves oxygen delivery
7	Maintain airway and assist breathing as needed	Prevents respiratory compromise
8	If symptomatic bradycardia persists → **Atropine 1 mg IV/IO** (repeat every 3–5 min, max 3 mg)	First-line chronotropic drug
9	If **atropine** ineffective → **Transcutaneous pacing**	Restores adequate heart rate
10	If **pacing** not effective → **Dopamine infusion (5–20 mcg/kg/min)**	Supports heart rate & BP
11	Alternatively or in addition → **Epinephrine infusion (2–10 mcg/min)**	Augments chronotropy and perfusion
12	Consider expert consultation / transvenous pacing if needed	Advanced pacing when initial measures fail
13	Identify and treat underlying causes (Hs & Ts)	Addresses reversible contributors
14	Reassess frequently, monitor response to therapy	Ensures appropriate care

Total Parenteral Nutrition (TPN)

- Definition of TPN
- Indications
- Components
- Route of administration
- Complications
- Nursing responsibilities
- Equipment
- Procedure



1. Definition

- **Total Parenteral Nutrition (TPN)** is the intravenous administration of **complete nutrition** to patients who cannot eat or absorb nutrients through the gastrointestinal (GI) tract.
- It provides all **macronutrients** (carbohydrates, protein, fat) and **micronutrients** (electrolytes, vitamins, trace elements).
- TPN bypasses the GI tract completely and is typically administered through a **central venous catheter**.
- Used **long-term or short-term**, depending on the condition.

2. Indications

TPN is indicated when the GI tract is non-functional or unsafe to use. Examples include:

- **Severe malnutrition**
- **Bowel obstruction**
- **Short bowel syndrome**
- **Severe pancreatitis**
- **Crohn's disease flare-up**
- **Peritonitis**
- **Severe burns or trauma**
- **Postoperative** conditions where enteral feeding is not possible
- **Cancer patients** undergoing chemotherapy or surgery
- **Coma or unconscious states**

3. Components of TPN

TPN solutions are customized to meet individual patient needs:

◆ *Macronutrients*

- **Carbohydrates**: Usually as **dextrose**; main energy source.

- **Proteins**: As **amino acids** for tissue repair and maintenance.
- **Fats**: As **lipid emulsions**, providing essential fatty acids and calories.

Micronutrients

- **Electrolytes**: Sodium, potassium, calcium, magnesium, phosphate, chloride.
- **Vitamins**: Water-soluble (e.g., B-complex, C) and fat-soluble (A, D, E, K).
- **Trace elements**: Zinc, copper, selenium, manganese, chromium, etc.

Fluids

- TPN also provides fluid for hydration and volume replacement.

4. Routes of Administration

- **Central venous catheter (CVC)** is most common (e.g., subclavian, internal jugular).
  - Necessary because high dextrose concentration is **irritating to peripheral veins**.
- **Peripheral Parenteral Nutrition (PPN)** is used for short-term or partial nutrition support.
  - Lower concentration to prevent vein irritation.

5. Administration Guidelines

- **Use aseptic technique** when handling TPN to reduce infection risk.
- TPN should be **infused via a dedicated lumen**—not shared with other IV medications.
- Start infusion slowly and **monitor reactions** (especially during first administration).
- **Tubing must be changed every 24 hours**.
- Never stop TPN abruptly—can cause **hypoglycemia**.
- Refrigerate TPN until use, then allow it to reach room temperature.

6. Monitoring Parameters

Daily or frequent monitoring is essential to prevent complications:

- **Vital signs** (especially temperature for infection)
- **Blood glucose** (risk of hyperglycemia)
- **Electrolytes**: Na^+, K^+, Ca^{++}, Mg^{++}, PO_4^{-}
- **Renal function**: BUN, creatinine
- **Liver function tests**: ALT, AST, ALP, bilirubin
- **Intake/output** and **daily weight**
- **Signs of infection** (catheter site and systemic)

7. Complications

Infectious

- Central line-associated bloodstream infection (CLABSI)

Metabolic

- Hyperglycemia or hypoglycemia
- Electrolyte imbalances (e.g., hypokalemia, hypophosphatemia)
- Refeeding syndrome (especially in malnourished patients)
- **Refeeding syndrome** is a potentially *life-threatening metabolic and electrolyte disturbance* that occurs when nutritional support (oral, enteral, or parenteral feeding) is initiated too quickly in malnourished or starved patients. It results from profound shifts in fluids and electrolytes during the transition from a catabolic (starving) to an anabolic (fed) metabolism
- Liver dysfunction (fatty liver, cholestasis)

Mechanical

- Catheter blockage, displacement, or thrombosis

- Air embolism

8. Nursing Responsibilities

- **Check TPN order** with pharmacy label—verify patient name, rate, contents.
- Inspect TPN bag for cloudiness, separation, or particles.
- **Prime tubing carefully**—avoid air bubbles.
- Monitor **blood glucose** and administer insulin if ordered.
- Keep central line **dressing clean and dry**; change per protocol.
- Educate patient/family on TPN purpose and care.
- Document infusion rate, patient tolerance, and any adverse reactions.

10. Weaning Off TPN

- Gradually reduce TPN rate while **reintroducing oral or enteral feeding**.
- Sudden stoppage can cause **hypoglycemia**.
- Monitor labs closely during the transition phase.
- Once the patient meets nutritional needs through the GI tract, TPN can be stopped.

5. Equipment Required

- TPN solution bag (custom-made in pharmacy)
- Infusion pump (to control rate accurately)
- Central venous access device (e.g., PICC line, tunneled catheter)
- IV tubing with **in-line filter** (to remove air and particles)
- Dressing supplies (for catheter site care)
- Blood sampling supplies (for lab monitoring)

TPN procedure checklist step with rationale

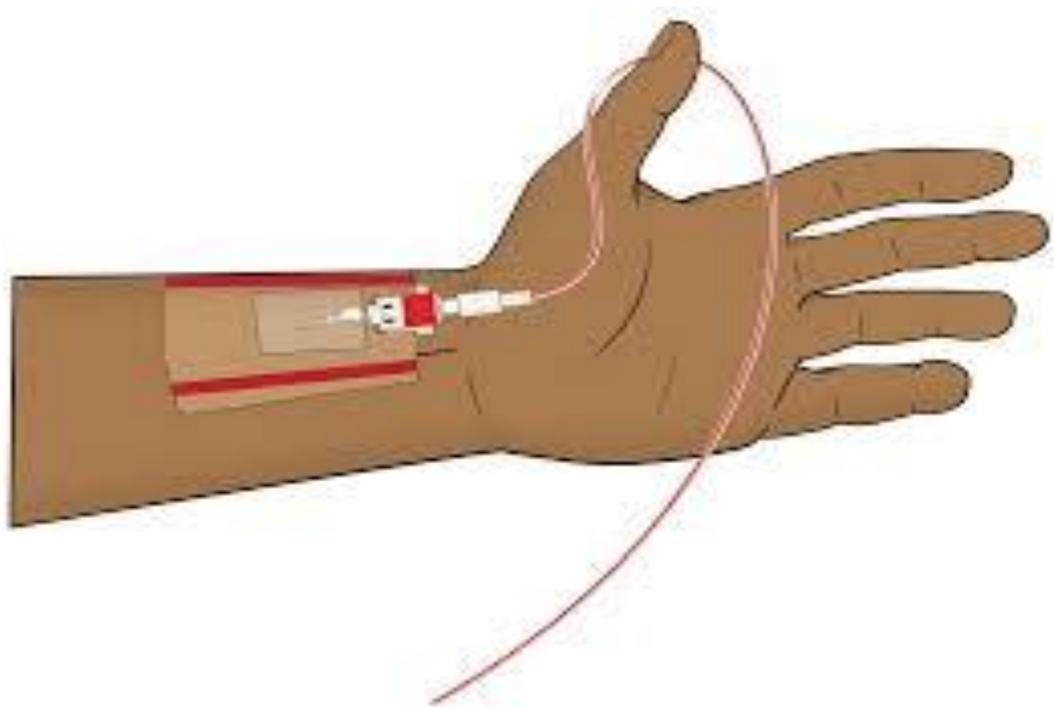
Step	Rationale
A. Nursing Preparation (Before TPN Administration)	
Verify the physician's TPN order. 	Ensures the correct formulation is given based on patient's daily needs.
Perform hand hygiene and don PPE.	Prevents infection during sterile procedures.
Assess patient's nutritional status and indication for TPN Review baseline laboratory investigations (CBC, electrolytes, RFT, LFT, **blood glucose**	• To confirm that the gastrointestinal tract cannot be used and that TPN is clinically indicated • Provides baseline values and helps identify electrolyte imbalances or organ dysfunction before initiating TPN.
Inspect the TPN bag for • Integrity • Clarity • label accuracy • Expiration .	Prevents administration of contaminated or expired solution.
Allow the TPN solution to warm to room temperature (if refrigerated).	Reduces discomfort and risk of vein irritation.
Assess the central line site and patency.	Ensure the line is functional and safe for hypertonic solution.
Label a dedicated lumen for TPN use only. Use a dedicated lumen for TPN infusion	Prevents incompatibility reactions and contamination.
Gather equipment needed**: infusion pump, IV tubing, inline filter, alcohol wipes.**	Prepares for efficient and sterile procedure.

Prime the tubing with the TPN solution using aseptic technique.	Prevents air embolism and maintains sterility.
B. Nursing Interventions (During and After TPN Administration)	
Connect tubing to central line using sterile technique.	Ensures safe and infection-free administration.
Start the infusion at the prescribed rate using an infusion pump. 	Delivers nutrients at a controlled and safe rate.
If discontinuing TPN, taper gradually	Prevents rebound hypoglycemia
Stop the infusion pump and clamp the line	Prevents air entry and blood reflux into the catheter.
Disconnect the TPN tubing using sterile technique	Maintains catheter sterility.
Flush the catheter with normal saline (usually 10–20 mL, per protocol)	Clears residual TPN from the line and maintains catheter patency
• Monitor blood glucose **after stopping TPN then** • Monitor blood glucose every **4–6** hours.	TPN can cause hyperglycemia or hypoglycemia.
• Monitor for complications: infection • fluid overload • electrolyte imbalance • refeeding syndrome.	Early detection reduces risk of life-threatening events.
Assess and document central line site every shift.	Prevents and detects catheter-related bloodstream infections.
Check and record intake/output and daily weight.	Monitors fluid balance and effectiveness of nutrition.
Review daily labs: electrolytes, liver/renal	Guides adjustments to TPN content and
function, glucose, triglycerides.	prevents complications.
Document all interventions, assessments, and patient responses.	Ensures accurate communication and legal accountability.
Provide patient/family education (if applicable).	Promote understanding and cooperation with therapy.

Arterial Catheter Care

Outlines:

- Definition
- Purposes & Indications
- Insertion Sites
- Arterial Pressure Waveform
- Complications
- Equipment
- Insertion & Care Procedure



Definition:

An arterial catheter is a thin tube inserted into an artery, primarily used in ICU and anesthesia for continuous blood pressure monitoring and arterial blood sampling.

Indications:

1. Frequent Blood Sampling:

- Respiratory conditions (ABG monitoring)
- Bleeding or metabolic abnormalities
- Monitoring serum levels during specific therapies

2. Continuous BP Monitoring:

- Hypotension, hypertension, shock types
- Use of vasopressors
- Mechanical cardiovascular support

Common Complications:

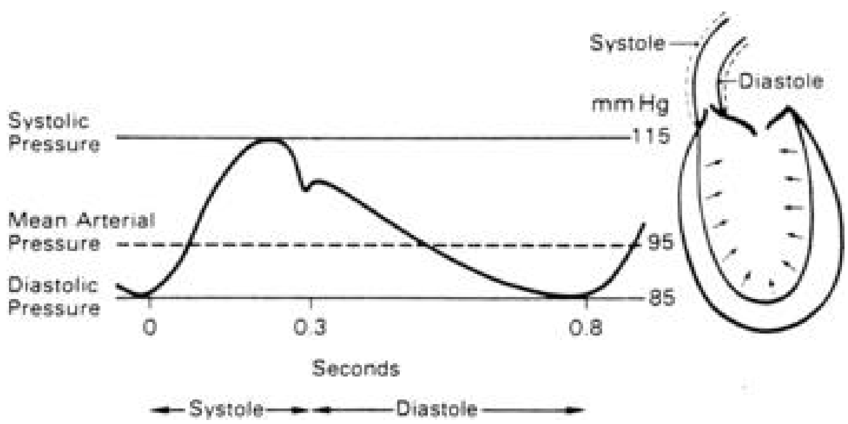
- Pain, vasospasm
- Hematoma, thrombosis, embolism
- Hemorrhage, ischemia, nerve injury
- Infection and vascular insufficiency
- Higher infection risk at femoral site

Sites of Insertion:

- **Radial Artery:** Most preferred; superficial and stable; requires modified Allen's test or Doppler confirmation.
- **Brachial Artery:** Deeper, without collateral circulation, higher risk.
- **Femoral Artery:** Used in emergencies; risk of hemorrhage, infection, and venous puncture.
- **Dorsalis Pedis & Posterior Tibial Arteries:** Least preferred due to high dislodgment

Arterial Pressure Waveform:

- Arterial pressure represents the forcible ejection of blood from the left ventricle into the aorta and out into the arterial system.
- During ventricular systole, blood is ejected into the aorta, generating a pressure wave. Because of the intermittent pumping action of the heart, this arterial pressure wave is generated in a pulsatile manner (See Figure).



- The ascending limb of the aortic pressure wave (anacrotic limb) represents an increase in pressure because of left-ventricular ejection.
- The peak of this ejection is the peak systolic pressure, which should be less than 120 mm Hg in adults.
- After reaching this peak, the ventricular pressure declines to a level below aortic pressure and the aortic valve closes, marking the end of ventricular systole.
- The closure of the aortic valve produces a small rebound wave that creates a notch known as the dicrotic notch.

- The descending limb of the curve (diastolic downslope) represents diastole and is characterized by a long declining pressure wave, during which the aortic wall recoils and propels blood into the arterial network.

- The diastolic pressure is measured as the lowest point of the diastolic down slope, which should be less than 80 mm Hg in adults.

The difference between the systolic and diastolic pressures is the pulse pressure, with a normal value of about 40 mm Hg.

- Arterial pressure is determined by the relationship between blood flow through the vessels (Cardiac Output) (CO) and the resistance of the vessel walls (Systemic Vascular Resistance) (SVR). The arterial pressure is therefore affected by any factors that change either cardiac output or systemic vascular resistance.

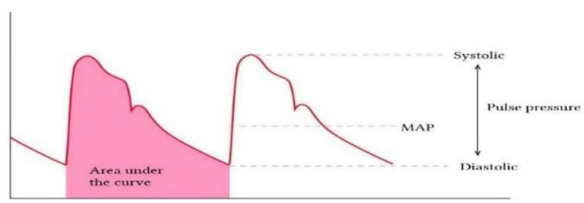
- The average arterial pressure during a cardiac cycle is called Mean Arterial Pressure (MAP).

- The MAP is calculated automatically by most patient monitoring systems; however, it can be calculated with the following formula:

MAP = (Systolic Pressure) + (Diastolic Pressure x 2)

- MAP represents the driving force (perfusion pressure) for blood flow through the cardiovascular system.

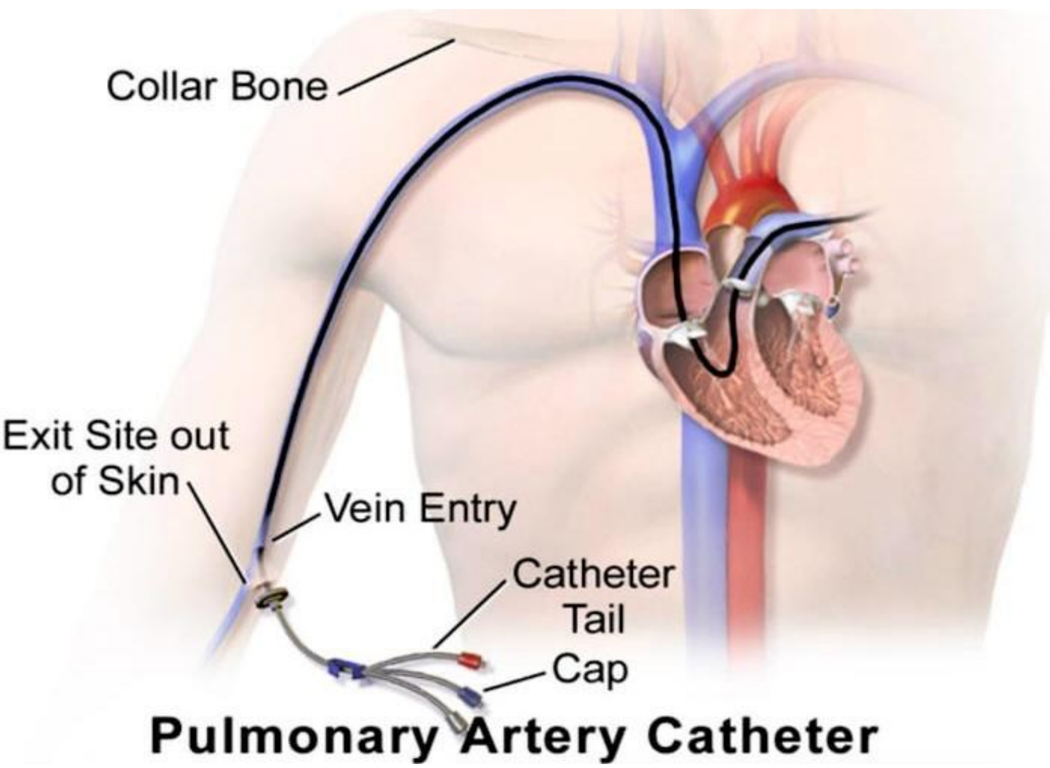
Procedure of arterial catheter care; -

N	Steps	Rationale
1	Verify physician order and explain procedure to patient.	Ensures correct treatment and reduces patient anxiety.
2	Perform hand hygiene and use aseptic technique.	Prevents infection.
3	Assess insertion site for redness, swelling, bleeding, or infection.	Early detection of complications.
4	Check arterial waveform and blood pressure accuracy.	Ensures proper functioning of arterial line. 
5	Maintain pressure bag at 300 mmHg.	Prevents blood backflow and clot formation.
6	Ensure tubing is free from air bubbles.	Prevents air embolism.
7	Zero and level transducer at phlebostatic axis.	Ensures accurate blood pressure readings.
8	Monitor circulation distal to insertion site (color, pulse, temperature).	Detects impaired circulation or ischemia.
9	Change dressing per hospital protocol using sterile technique.	Reduces risk of infection.
10	Document assessments, readings, and patient response.	Provides legal record and continuity of care.

Pulmonary catheter care

Outlines: -

- Definition
- Indications
- Contraindications
- Complications
- Mechanism of work
- Equipment
- Procedure



Definition

A pulmonary artery catheter is a balloon-tipped, flow-directed catheter inserted through a central vein and advanced into the pulmonary artery to measure cardiac and pulmonary hemodynamic parameters such as pulmonary artery pressure, pulmonary capillary wedge pressure, and cardiac output.

Indications

- Hemodynamic monitoring in critically ill patients
- Shock (cardiogenic, septic, hypovolemic)
- Severe heart failure
- Acute myocardial infarction with complications
- ARDS
- Major cardiac or thoracic surgery
- Severe pulmonary hypertension
- Assessment of fluid status and cardiac output

Contraindications

Absolute Contraindications:

- Pulmonary artery rupture
- Right atrial or ventricular tumor
- Infection at insertion site

Relative Contraindications:

- Coagulopathy
- Severe arrhythmias
- Pulmonary hypertension
- Tricuspid or pulmonary valve disease

Complications

- Cardiac arrhythmias
- Pulmonary artery rupture or infarction
- Thrombosis and embolism
- Infection (catheter-related bloodstream infection)
- Pneumothorax during insertion
- Air embolism
- Balloon rupture or catheter knotting

Physiology of Work (Mechanism of Action)

The pulmonary artery catheter is inserted through a central vein and advanced sequentially into the right atrium, right ventricle, and pulmonary artery. Balloon inflation allows the catheter to float with blood flow. Pressure transducers measure intracardiac and pulmonary pressures. Pulmonary capillary wedge pressure reflects left ventricular preload. Cardiac output is measured using the thermodilution technique.

Normal range:

1. Right Atrial Pressure (RAP) / CVP:
   (2-8 mmHg)
2. Right Ventricular Pressure (RVP):
   Systolic: 15-30 mmHg
   Diastolic: 2-8 mmHg
3. Pulmonary Artery Pressure (PAP):
   Systolic: 15-30 mmHg
   Diastolic: 4-12 mmHg
   Mean (mPAP): 10-20 mmHg

Equipment Needed

- Pulmonary artery catheter kit
- Sterile gloves and gown
- Chlorhexidine antiseptic solution
- Sterile transparent dressing
- Pressure transducer and monitoring system
- Normal saline or heparinized saline
- Syringe (1–1.5 mL) for balloon inflation
- ECG and hemodynamic monitor
- Sterile gauze and waste disposal bag

Procedure: -

Step	Procedure	Rationale
1	Perform hand hygiene and apply sterile gloves	Prevents cross infection
2	Assess catheter insertion site	Detects early signs of infection or bleeding 
3	Ensure catheter is secured properly	Prevents displacement or migration
4	Clean insertion site with chlorhexidine	Reduces microbial contamination
5	Apply sterile transparent dressing	Protects site and allows observation
6	Level transducer at phlebostatic axis	Ensures accurate pressure measurement 
7	Zero the transducer system	Eliminates atmospheric pressure effects

8	Monitor pulmonary artery pressures	Assesses hemodynamic status 
9	Inflate balloon only during PCWP measurement	Prevents pulmonary artery damage
10	Deflate balloon immediately after reading	Restores normal blood flow
11	Flush catheter lumens as prescribed	Maintains catheter patency
12	Monitor ECG and vital signs continuously	Detects arrhythmias and instability
13	Document all findings and patient response	Ensures continuity of nursing care

ICU Scoring Systems and Assessment Scales

Outlines: -

- **Glasgow Coma Scale (GCS)**
- **Pain score**
- **Fall Risk Assessment Scales**
- **Braden Scale (Pressure Ulcer Risk)**
- **Richmond Agitation-Sedation Scale (RASS)**

1. Glasgow Coma Scale (GCS)

Component	Response	core
Eye Opening (E)	Spontaneous	4
	To voice	3
	To pain	2
	None	1
Verbal Response (V)	Oriented	5
	Confused	4
	Inappropriate words	3
	Incomprehensible sounds	2
	None	1
Motor Response (M)	Obeys commands	6
	Localizes to pain	5
	Withdrawal from pain	4
	Flexion to pain (decorticate)	3
	Extension to pain (decerebrate)	2
	None	1

Total Score: 3-15 (Severe: 3-8, Moderate: 9-12, Mild: 13-15)

2. Pain Assessment Scales

A. Numeric Rating Scale (NRS)

Score	Description
0	No pain
1-3	Mild pain
4-6	Moderate pain
7-9	Severe pain
10	Worst possible pain

B. Wong-Baker FACES Pain Rating Scale

Face	Score	Description
😊	0	No hurt
🙂	2	Hurts little bit
😐	4	Hurts little more
😟	6	Hurts even more
😢	8	Hurts whole lot
😭	10	Hurts worst

C. BPS (Behavioral Pain Scale)

Item	Description	Score
Facial Expression	Relaxed	1
	Partially tightened	2
	Fully tightened	3
	Grimacing	4
Upper Limbs	No movement	1
	Partially bent	2
	Fully bent with finger flexion	3
	Permanently retracted	4
Compliance with Ventilation	Tolerating movement	1
	Coughing but tolerating	2
	Fighting ventilator	3
	Unable to control ventilation	4

Total Score: 3-12 (>5 indicates unacceptable pain)

3. Fall Risk Assessment Scales

A. Morse Fall Scale

Risk Factor	Scale	Score
History of falling	No	0
	Yes	25
Secondary diagnosis	No	0
	Yes	15
Ambulatory aid	None/bedrest/nurse assist	0
	Crutches/cane/walker	15
	Furniture	30
IV/Heparin Lock	No	0
	Yes	20
Gait/Transferring	Normal/bedrest/immobile	0
	Weak	10
	Impaired	20
Mental Status	Oriented to own ability	0
	Forgets limitations	15

Total Score: Low Risk: 0-24, Moderate Risk: 25-50, High Risk: ≥51

4. Braden Scale (Pressure Ulcer Risk)

Parameter	1	2	3	4
Sensory Perception	Completely limited	Very limited	Slightly limited	No impairment
Moisture	Constantly moist	Very moist	Occasionally moist	Rarely moist
Activity	Bedfast	Chairfast	Walks occasionally	Walks frequently
Mobility	Completely immobile	Very limited	Slightly limited	No limitations
Nutrition	Very poor	Probably inadequate	Adequate	Excellent
Friction & Shear	Problem	Potential problem	No apparent problem	N/A

Total Score: 6-23

- Very High Risk: ≤9
- High Risk: 10-12
- Moderate Risk: 13-14
- Mild Risk: 15-18
- No Risk: 19-23

Detailed Braden Scale Descriptions

Sensory Perception:

- **1 - Completely Limited:** Unresponsive to painful stimuli
- **2 - Very Limited:** Responds only to painful stimuli, cannot communicate discomfort
- **3 - Slightly Limited:** Responds to verbal commands, has some sensory impairment
- **4 - No Impairment:** Responds to verbal commands, has no sensory deficit

Moisture:

- **1 - Constantly Moist:** Skin kept moist almost constantly
- **2 - Very Moist:** Skin often but not always moist
- **3 - Occasionally Moist:** Skin occasionally moist
- **4 - Rarely Moist:** Skin usually dry

Activity:

- **1 - Bedfast:** Confined to bed
- **2 - Chairfast:** Ability to walk severely limited
- **3 - Walks Occasionally:** Walks occasionally during day
- **4 - Walks Frequently:** Walks outside room at least twice daily

Mobility:

- **1 - Completely Immobile:** Does not make even slight changes in body/extremity position
- **2 - Very Limited:** Makes occasional slight changes in body/extremity position
- **3 - Slightly Limited:** Makes frequent though slight changes in position
- **4 - No Limitations:** Makes major/frequent changes in position

Nutrition:

- **1 - Very Poor:** Never eats complete meal, rarely eats more than 1/3
- **2 - Probably Inadequate:** Rarely eats complete meal, generally eats 1/2
- **3 - Adequate:** Eats over half of most meals
- **4 - Excellent:** Eats most of every meal, never refuses meal

Friction & Shear:

- **1 - Problem:** Requires moderate to maximum assistance, complete lifting impossible
- **2 - Potential Problem:** Moves feebly, requires minimum assistance
- **3 - No Apparent Problem:** Moves independently, maintains good position in bed/chair

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Richmond Agitation-Sedation Scale (RASS)

Score	Term	Description
+4	Combative	Overtly combative, violent, immediate danger to staff
+3	Very agitated	Pulls/removes tubes or catheters, aggressive
+2	Agitated	Frequent non-purposeful movement, fights ventilator
+1	Restless	Anxious, apprehensive, movements not aggressive
0	Alert and calm	
-1	Drowsy	Not fully alert, sustained awakening to voice (≥10 sec)
-2	Light sedation	Briefly awakens to voice (<10 sec)
-3	Moderate sedation	Movement or eye opening to voice, no eye contact
-4	Deep sedation	No response to voice, movement or eye opening to physical stimulation
-5	Unarousable	No response to voice or physical stimulation

Note: These scales should be used as clinical tools in conjunction with comprehensive patient assessment and clinical judgment.

Bundles of Care of Critically ill patients

Ventilator associated pneumonia (VAP) bundles

Catheter associated urinary tract infection (CAUTI) bundles

Center line associated blood stream infection CLABSI Bundles

📋 VAP Bundle (Ventilator-Associated Pneumonia)

Daily Assessment Schedule

Time	Intervention	Rationale
Every Shift	Assess readiness for extubation	Early liberation reduces VAP risk
Every Shift	Elevate head of bed 30-45°	Prevents aspiration of secretions
Every Shift	Assess sedation level & daily sedation vacation	Reduces ventilator days
Every 2-4 Hours	Oral care with chlorhexidine 0.12%	Reduces bacterial colonization
Every 4 Hours	Suction only when needed (not routine)	Prevents trauma and colonization
Every Shift	Check ET tube cuff pressure (20-30 cm H_2O)	Prevents aspiration
Daily	Spontaneous breathing trial (if appropriate)	Assesses extubation readiness
Every Shift	DVT prophylaxis	Prevents complications
Every Shift	Peptic ulcer prophylaxis	Prevents stress ulcers

VAP Bundle Checklist

- ✅ HOB elevated 30-45°
- ✅ Daily sedation vacation/assessment
- ✅ Daily extubation assessment
- ✅ Oral care with antiseptic
- ✅ DVT prophylaxis
- ✅ Peptic ulcer prophylaxis
- ✅ Verify ET tube position and cuff pressure

CAUTI Bundle (Catheter-Associated Urinary Tract Infection)

Daily Assessment Schedule

Time	Intervention	Rationale
At Insertion	Use sterile technique for insertion	Prevents initial contamination
At Insertion	Use smallest appropriate catheter size	Reduces urethral trauma
Every Shift	Assess continued need for catheter	Early removal reduces infection risk
Every Shift	Maintain closed drainage system	Prevents bacterial entry
Every Shift	Keep drainage bag below bladder level	Prevents backflow of urine
Every Shift	Secure catheter to prevent movement	Reduces urethral trauma
Every Shift	Perform perineal hygiene	Maintains cleanliness
Every Shift	Empty drainage bag regularly (do not touch spout)	Prevents bacterial growth
Daily	Document indication for catheter	Ensures appropriate use

CAUTI Bundle Checklist

- ✅ Catheter only if indicated (appropriate use)
- ✅ Sterile insertion technique used
- ✅ Maintain closed drainage system
- ✅ Bag below bladder level always
- ✅ Catheter secured properly
- ✅ Perineal care performed
- ✅ Daily assessment for removal

💉 CLABSI Bundle (Central Line-Associated Bloodstream Infection)

Time	Intervention	Rationale
At Insertion	Use maximal sterile barrier precautions	Prevents contamination during placement
At Insertion	Use chlorhexidine skin antisepsis	Superior antiseptic for skin prep
At Insertion	Select optimal catheter site (avoid femoral)	Subclavian preferred, then IJ
Every Shift	Assess continued need for central line	Early removal reduces infection risk
Every Shift	Perform hand hygiene before accessing line	Prevents pathogen transmission
Every Shift	Inspect insertion site	Early detection of complications
Every Shift	Scrub the hub 15 seconds before access	Reduces microbial contamination
Every 5-7 Days	Change transparent dressing	Maintains sterile environment
Every 24 Hours	Change gauze dressing	If used instead of transparent
Every Shift	Change needleless connectors per protocol	Reduces biofilm formation
Every 24-96 Hours	Change IV tubing per protocol	Prevents bacterial growth
Daily	Review labs for signs of infection	Early detection of CLABSI

- ✅ Hand hygiene performed
- ✅ Maximal sterile barriers during insertion
- ✅ Chlorhexidine skin prep used (>0.5%)
- ✅ Optimal site selection (avoid femoral)
- ✅ Daily review of line necessity
- ✅ Dressing clean, dry, and intact
- ✅ Hub scrubbed 15 seconds before access
- ✅ Needleless connectors changed per policy
- ✅ No signs of infection at site

ICU Medications

Medication	Dose	Action	Indications	Nursing Role
Propofol (Diprivan)	5-50 mcg/kg/min IV	Sedates brain quickly	Mechanical ventilation, procedures, seizures	• Monitor BP (causes hypotension) • Check lipids (milky white) • Change tubing q12h • Wake up test daily
Midazolam (Versed)	0.02-0.1 mg/kg/hr IV	Short-acting sedative, causes amnesia	Sedation, seizures, anxiety	• Monitor respiratory rate • Have reversal ready (Flumazenil) • Assess sedation level • Can cause tolerance
Fentanyl	25-200 mcg/hr IV	Powerful pain relief	Pain, sedation on vent	• Monitor respiratory depression • Check BP and HR • Have Narcan ready • Assess pain level
Dexmedetomidine (Precedex)	0.2-0.7 mcg/kg/hr IV	Sedates without stopping breathing	Sedation allows patients to wake easier	• Monitor HR (causes.bradycardia) • Check BP • Patient can be awake but calm • Good for weaning

VASOPRESSORS (Blood Pressure Support)

Medication	Dose	Action	Indications	Nursing Role
Norepinephrine (Levophed)	0.01-3 mcg/kg/min IV	Squeezes blood vessels tight → raises BP	Septic shock, low BP	• Central line only • Monitor BP q5-15min • Check extremities (can cause necrosis) • Titrate to MAP >65
Epinephrine (Adrenaline)	0.01-0.5 mcg/kg/min IV	Stronger heart + tight vessels	Cardiac arrest, anaphylaxis, severe shock	• Monitor HR (increases) • Watch for arrhythmias • Check glucose (increases) • Central line only
Dopamine	2-20 mcg/kg/min IV	Low dose: kidneys work High dose: raises BP	Shock, low urine output	• Dose-dependent effects • Monitor HR • Central line preferred • Less used now
Dobutamine	2-20 mcg/kg/min IVMakes	heart squeeze strongerMonitor HR (increases)	Heart failure, cardiogenic shock•	• Check BP (may drop) • Watch rhythm (arrhythmias) • Assess cardiac output

ANTICOAGULANTS

Medication	Dose	Action	Indications	Nursing Role
Heparin	80 units/kg bolus then 18 units/kg/hr IV	Prevents blood clots	DVT, PE, stroke, heart attack	• Monitor aPTT q6h • Check platelets (HIT) • Watch for bleeding • Have protamine ready
Enoxaparin (Clexan)	1 mg/kg SC q12h	Prevents blood clots (longer acting)	DVT prevention/treatment	• Give in abdomen • Don't rub site • Watch for bleeding • Adjust for renal function

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